The Morphosyntax of Hasawi Arabic's Tense, Aspect, and Aktionsart

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Abstract

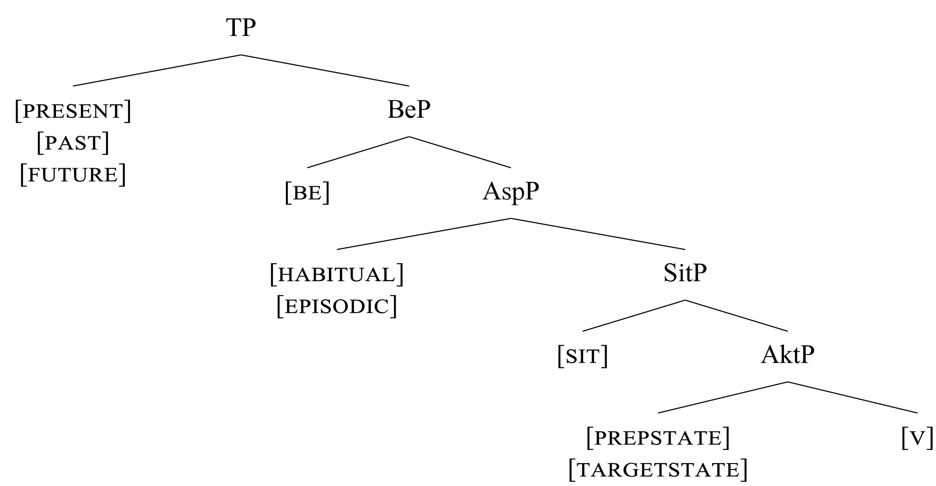
This paper claims on the basis of Hasawi Arabic's morphosyntax that the habitual aspect, and potentially other categories such as tense, durative aspect (prolonged eventuality), and episodic aspect (a single eventuality, in contrast to the habitual) are structurally higher than the aktionsart categories such as preparatory and target states (the difference between "to fall asleep" and "to sleep", respectively). This proposed structure is realized by a mixture of conjugational morphology (the conjugational paradigms of prefixal conjugation, suffixal conjugation, and poor suffixal conjugation), auxiliary roots such as $\sqrt{\text{gʕd}}$ (a lexically bleached root of "sit") and $\sqrt{kwn}$ "be", and lexical roots. I contend that these roots and conjugations can be compositionally put together in order to yield a systematic analysis of the morpho-syntax-semantics of Hasawi Arabic's auxiliary system.

Contents

1 Introduction

Hasawi Arabic (HA) is an understudied dialect of Arabic spoken in the Eastern Province of Saudi Arabia. Based on verbal conjugations and roots to be introduced, I contend that HA has the following syntactic structure:

(1)



In (1), the heads [V], [SIT], and [BE] all occupy their respective terminal nodes. The other terminal nodes are targets of competition by multiple heads: [TARGETSTATE] and [PREP(ARATORY)STATE] compete for the terminal node of the Akt(sionart)P, [EPISODIC] and [HABITUAL] for the terminal node of Asp(ect)P, and [PRESENT], [PAST], and [FUTURE] for the terminal node of TP. Throughout this paper, I will provide evidence for the hierarchy of the structure in (1), as well as clarify more on the semantics of these heads, but before I do that, I need to digress and introduce facts about the verbal morphology of Arabic in order to explain how these heads are realized.

In Arabic (Classical or the modern dialects), the verb consists of the following elements: (i) the lexical or auxiliary root (usually consisting of trilateral consonants, such as $\sqrt{ktb}$ "write"). (ii) a CV-skeletal structure (called "form", "measure", or "binyan" [1] in the literature) that usually gives information about argument structure such as (in)transitivity, reflexivity, and reciprocality, or some aspectual information such as the intensity of an event or its pluractionality. (iii) the content (phonological quality) of the vowel slots of the skeletal structure which denotes active/passive voice[2]. (iv) finally, agreement morphology, which is sensitive to tense, aspect, and aktionsart (the conjugational paradigms). To illustrate these four elements, consider sentences (2)-(3) from Classical Arabic[3] which all have the same root $\sqrt{\text{ksɹ}}$ "break"[4]:

(2) a. kasaɹ-a l-walad-u l-baːb-a
break-3.SG.M DEF-boy-NOM DEF-door-ACC
"The boy broke the door."

b. kusiɹ-a l-baːb-u
break.PASS-3.SG.M DEF-door-NOM
"The door was broken."

(3) a. kasːaɹ-a l-walad-u l-baːb-a
break.INT-3.SG.M DEF-boy-NOM DEF-door-ACC
"The boy broke the door (to pieces)."

b. kusːiɹ-a l-baːb-u
break.INT.PASS-3.SG.M DEF-door-NOM
"The door was broken (to pieces)."

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[1] See Wright (1996, P29), McCarthy (1981), Guerssel and Lowenstamm (1996), Fassi Fehri (2001). Throughout this paper, I will reserve the term "form" specifically for this CV-skeleton.

[2] In Classical Arabic's Form I, vowel quality might denote things other than voice, such as change of state in a subset of verbs. See Guerssel and Lowenstamm (1996) for a detailed examination of Form I vowel qualities and alternations.

[3] The logic in the sentences (2)-(3) applies to modern dialects of Arabic, with slight variation. Fassi Fehri (1993) notes that most dialects of Arabic lost the passive voice alternations with the so called "perfect" conjugation. Instead of the stem CaCaC alternating to CuCiC (or vice versa), prefixes such as /n-/ or /t-/ are used to signal passive voice, which gives the result of CaCaC > n/tCaCaC. There are minor vestiges of such vowel alternations for the passive in HA. Similar to Fassi Fehri claims, HA requires the affixes /n-/ and /t-/ in passives, but the stem vowels also show change. Compare the verb [jiguul] "He says" with its passive version [jingaal] "It is said", where in addition to the prefix /n-/ the stem changed from [guul] to [gaal].

[4] All of the Arabic examples in this paper are written with the IPA. The glossings are 1 = first person, 3 = third person, ACC = accusative, DAT = dative, DEF = definite, DEM = demonstrative, F = feminine, FUT = future, IMP = imperative, INS = instrumental, INT = intensive aspect, M = masculine, NOM = nominative, PASS = passive, PC = prefixal conjugation, PL = plural, PROG = progressive, PSC = poor suffixal conjugation, SC = suffixal conjugation, SG = singular.

The following morphophonological explanation is based off McCarthy (1981). In (2a) and (2b), the triliteral root $\sqrt{\text{ksɹ}}$ is associated to what is called Form I, which has the CV-skeleton /CVCVC/. Semantically, Form I is underspecified in terms of the intensity of the event. In sentences (3a) and (3b) on the other hand, the root is associated to what is called Form II, which has the CV-skeleton /CVCCVC/ and which entails intensity of the event (at least for a subset of roots such as $\sqrt{\text{ksɹ}}$). We can see here that the entailment of the intensity of the event or its underspecification depends on what form the root $\sqrt{\text{ksɹ}}$ is associated to.

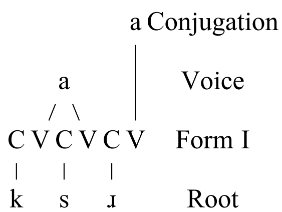
As for voice, we can see that it is sensitive to the qualities of the vowels. (2a) has the stem /CaCaC/, and (3a) has the stem /CaCCaC/, both sentences have all of their vowel slots with the qualities of the vowel /a/, and both are in the active voice. On the other hand, (2b) has the stem /CuCiC/, and (3b) /CuCCiC/, both sentences have an initial vowel slot with the qualities of the vowel /u/, and a second V slot with the qualities of the vowel /i/, and both are in the passive voice. It is safe to say that the /...a...a.../ pattern realizes the active voice, while the /...u...i.../ pattern realizes passive voice.

Lastly, and more importantly, in all of these sentences, the verb is in the "perfect"/"past" conjugation, which denotes a simple past reading (at least in these sentences). The marker of this conjugational paradigm is the third person masculine singular morpheme /-a/. This agreement morpheme only appears in this conjugation and nowhere else.

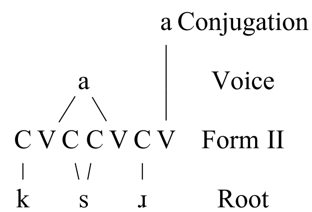
As we have seen, verbal "forms" and vowel quality have nothing to do with the past tense in (2) and (3): Whether the root is associated to Form I or Form II, or whether all vowel slots have /a/s or not, the sentences yield past tense readings. The key factor here for tense is the agreement morphology which reflects the conjugational paradigm. Because of this, I will ignore the verbal forms and vowel qualities in this paper. I will instead focus on the roots (lexical and auxiliary) and the conjugational paradigms, as they are the morphemes that realize the heads in (1), as I will show[5].

[5] Some linguists working on Classical Arabic (or the dialect with no clear native speakers "Modern Standard Arabic") might disagree with my analysis of agreement morphology essentially being portmanteau with tense, aspect, and aktionsart. They instead consider these morphemes purely agreement morphemes, and instead consider "ablaut" such as the change of the stem vowel in "perfective" [kataba] "He wrote" to "imperfective" [jaktub] "He writes" to be the marker of tense/aspect. This in my view is false. To clearly see this, consider Form V. This form has the exact same stem in the "perfective" as well as "imperfective" conjugations. Taking the root $\sqrt{\text{qbl}}$ "to receive/accept (a gift)" as an example, compare perfective Form V [taqabːal-a] "He accepted/received (a gift)" and "imperfective" Form V [ja-taqabːal] "He accepts/receives (a gift)". In both conjugations, the stem is exactly the same, namely /taCaCCaC/, the only differentiating factor are agreement morphemes, which clearly show past for the former and some sort of generecity/habituality to the latter. One can take these active verbs and turn them into passives by changing the quality of their vowels. Thus [taqabːal-a] becomes [tuqubːil-a] "It was accepted/received" and [ja-taqabːal] becomes [ju-taqabːal] "It is accepted/received". In all of the former cases of active and passive verbs, it is very clear that the constant factor for past tense is the agreement suffix /-a/, and the constant factor for the generic/habitual aspect is the agreement prefix /jV-/. Thus I stand by my claim that vowel qualities and its alternations have nothing to do with tense, aspect or aktionsart: Agreement morphemes in Arabic (Classical and Hasawi Arabic) are portmanteau morphemes that combine agreement and the aforementioned categories.

(4)



(5)



Using McCarthy (1981)'s theory of Autosegmental (Morpho)phonology, we can represent how the aformentioned morphemes of Classical Arabic are associated: The root $\sqrt{\text{ksɹ}}$ is a short-hand way of representing the subsegmental features ([+ Plosive], [+ Voiceless], etc.), which associate to the CV-skeleton of the forms. The forms also have their V slots associated with the subsegmental features of vowels that realize voice. Finally, tense/aspect/aktionsart and agreement portmanteau morphemes affix on this CV-skeleton. (4) illustrates the verb in sentence (2a) and (5) illustrates the verb in sentence (3a). For how these morphemes are syntactically put together, see the tree in (6) below. I will now focus on the conjugations.

Arabic, as well as a subset of Semitic languages, have four conjugational paradigms. To illustrate them and the agreement morphemes that occur with them, I present Table 1 that shows HA's conjugational paradigms with the root $\sqrt{\text{ktb}}$ "write"[6].

Imperative Conjugation			Poor Suffixal Conjugation		
	SG	PL		SG	PL
F	kɪtbæj	kɪtbaw	F	kaːtbah	kaːtbiːn
M	ʔɪktəb		M	kaːtəb	
Suffixal Conjugation			Prefixal Conjugation		
	SG	PL		SG	PL
1	ktabt	ktabna	1	ʔaktəb	naktəb
2F	ktabtæj	ktabtaw	2F	taktəbiːn	taktəbuːn
2M	ktabt		2M	taktəb	
3F	ktabat	ktabaw	3F		jaktəbuːn
3M	kətab		3M	jaktəb	

Table 1: HA Verbal Conjugations for $\sqrt{\text{ktb}}$ "write"

Following Shlonsky (1997)'s terminology for Modern Hebrew, I will adopt a semantics-neutral labelling: Prefixal Conjugation (PC), Suffixal Conjugation (SC), and Poor Suffixal Conjugation (PSC)[7]. In the PC, person is realized by a prefix, while gender and number are realized by suffixes (except for the 1st plural, which appears to be a portmanteau prefix of person and number). In the SC, all phi features are realized by suffixes. Finally, the PSC is similar to the SC, but it is "poor" as it only has gender and number as suffixes, but no person distinction.

[6] I only mention the imperative conjugation here for the sake of giving the full picture of the conjugational paradigm. Otherwise, it is orthogonal to the syntactic structure in (1).

[7] Shlonsky (1997) calls this the "Benoni Form", meaning "the intermediate", as it could be used for verbs and nouns.

In Arabic's grammatical and linguistic literature, these conjugational paradigms are labelled variously with quasi-semantic labellings that are inappropriate. The PC is labelled as the "imperfect", "present", "imperfective", or "non-past", the SC as "perfect", "perfective", or "past", and the PSC as the "(active) participle". An example of why these labels are inappropriate, if not outright misleading, is the fact that in some dialects such as HA, as well as Najdi Arabic[8], the so called "participle" is used for perfect readings (event time preceding relative time in terms of Reichenbach (1947)). It is more appropriate to think of these paradigms as having a one to many relationship between morphology and semantics: The SC can denote past or perfect semantics in certain morphosyntactic contexts, but not always. The exception to this is the imperative, as it indeed corresponds to a one to one relationship between the conjugation and the imperative meaning.

Let us return to the heads in (1) and how they are realized by roots and conjugational paradigms. Following a similar approach to Amberber (2010)'s account of Amharic, I take the heads [V], [SIT], and [BE] to be roots that underwent head movement to heads (such as V and v) realized as CV-skeletal forms, as in (6)[9]. The result of this movement is the formation of verbal stems. [V] is a stand alone for any root that carries lexical meaning (for example, $\sqrt{ktb}$ "write"). [SIT] is realized by the auxiliary root $\sqrt{\text{gʕd}}$ "sit", and [BE] is realized by the auxiliary root $\sqrt{\text{kwn}}$ "be". I call the latter two roots "auxiliaries" as they do not contribute any lexical meaning (no real "sitting" is inolved with the auxiliary $\sqrt{\text{gʕd}}$, see §4). Other authors differ in naming these verbs. The literature include labels such as "light verbs", "serial verb constructions", or "bleached verbs", among others[10]. For the purposes of this paper, I only draw a distinction between the roots that carry lexical meaning and those that don't. Thus I will call the former "lexical roots" and the latter "auxiliary roots". If the need to address their stems arise, I will use "lexical stems" and "auxiliary stems".

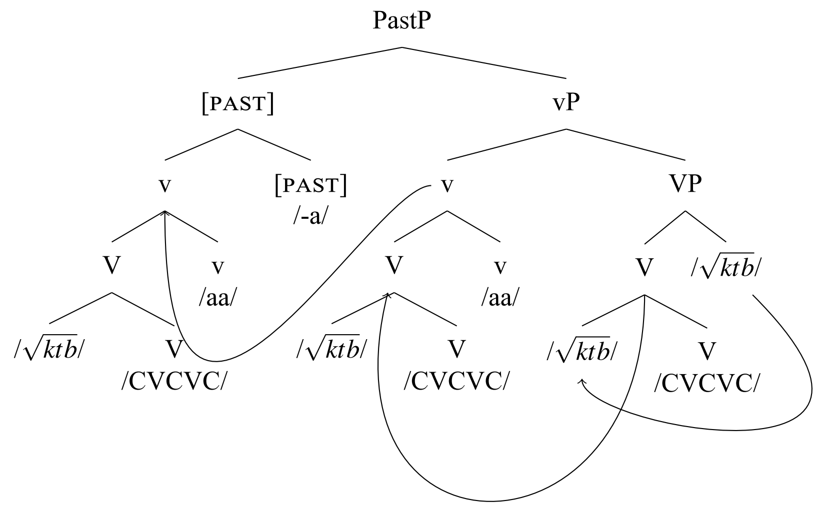
As for the other heads, they are realized by conjugational morphology. For example, as we have seen in (2)-(3), the verbal stems are affixed with the SC that realizes the head [PAST] via an agreement morpheme /-a/. Following Fassi Fehri (1993), I assume that the verbal stem in Arabic undergoes head-movement from V to I (in current terms, V to [PAST]), as in (6), which is a syntactic derivation for the verb [kataba] "He wrote". The root undergoes head-movement to V which creates the verbal form (Form I, in this case). Then, $\sqrt{ktb}+\text{V}$ undergoes head-movement to v (which realizes the active morphology, as the quality of the vowels is /CaCaC/). Lastly, the complex head $\sqrt{ktb}+\text{V}+\text{v}$ undergoes head-movement to [PAST].

[8] I am grateful to Khawlah Allahim for this information.

[9] While it is understandable that a lexical root may undergo movement to V and then little v, an elaboration is required when auxiliary roots do the same. Evidence that auxiliary stems have a similar structure to lexical stems is the fact that they appear with the CV-skeleton of the forms. However, the morphology of the forms on the auxiliary roots contributes nothing to the semantics of argument structure or aspectual intensity/pluractionality. Amberber (2010, P311, fn.7) also notes this in Amharic, where both the lexical stem and the auxiliary stem seem to be "complex" (both containing form morphology). That the form morphology on the auxiliary stems does not contribute to meaning suggests that it could be some sort of default morphology. Most auxiliary roots appear in Form I. Take for example the auxiliary roots $\sqrt{\text{gʕd}}$ and $\sqrt{\text{kwn}}$ which only link to Form I. Compare this with HA's lexical root $\sqrt{\text{gʕd}}$ which can appear in Form I as gəʕad "He sat" and Form II gaʕːad "He caused X to sit".

[10] See Ouali and Al Bukhari (2016) and the resources there for other posture verbs in other Arabic dialects.

(6)



(6) is simplified with various respects. First, similar to (1), I am ignoring here argument structure, as it is orthogonal to the realization of the syntactic heads in (1). It does not matter whether we have di-transitive, mono-transitive, or intransitive sentences, as the SC will appear on the verbal stem because of the head [PAST]. I will also not attempt to account for how agreement or phi features interact with heads such as [PAST]. For agreement, one could plug in one's favourite theory of agreement. As for phi features themselves, it does not matter what combination of number, gender, or person there is in the sentence. What matters is that the head [PAST] creates an environment on the verbal stem that has a one to one relationship between it and the set of agreement morphemes in the suffixal conjugation paradigm of Table 1[11][12]. This logic extends to the other heads, with the heads [FUTURE], [HABITUAL], and [PREPSTATE] being realized by the PC (the set of agreement morphemes of the PC) and the heads [EPISODIC] and [TARGETSTATE] being realized by the PSC (the set of agreement morphemes of the PSC). Finally, I take the head [PRESENT] to be phonologically null (see §3.3 for justifications).

The combination of roots and conjugational morphology will be the evidence for the structure in (1). In (6), there is only one root, the lexical root $\sqrt{ktb}$, and one conjugational paradigm, the SC, realizing the head [PAST]. However, as I will show later, once we introduce auxiliary roots alongside the lexical root, each root will be associated to a CV-skeleton and create a stem. Then, each verbal stem will be affixed with conjugational morphology, giving evidence to (1), as when there are multiple

[11] I have used a Classical Arabic example in (6), even though Table 1 is based off my Hasawi Arabic. This is because it is easier to show agreement morphology in Classical Arabic (the morpheme /-a/), as Hasawi Arabic has some morphophonological nuances that I do not want to get into. Otherwise nothing changes with the presented logic.

[12] In Sason Arabic, phi features control the appearance of a copular verb (and by extension, the appearance of a conjugation on said verb), Akkuş and Benmamoun (2016, P164). This never happens in HA.

stems in the same sentence, there are restrictions on which stem the conjugational morphology attach to.

The following sections will try to justify the syntactic hierarchy of the heads above. In §2, I will introduce a puzzle of selectional restriction on lexical roots. I do so in order to introduce the aspectotemporal structure which is the reason for postulating the aktionsart heads [PREPSTATE] and [TARGETSTATE] and which I take to be the puzzle's solution. After explaining the puzzle and justifying the features [PREPSTATE] and [TARGETSTATE], §3 shows how all the heads that are realized as conjugations can affix to lexical stems only, without any auxiliary roots/stems being present. It also shows that when there are multiple conjugations and not enough verbal stems, ”be-support” activates, and the auxiliary root $\sqrt{\text{kwn}}$ “be” is inserted in order to support the morphology of the ”overflowing” conjugations (see Bjorkman (2011, P27-28) for such claims for Arabic and other languages). After justifying the syntactic heads and giving evidence on their independence, §4 explores the nature and the semantic contribution of the auxiliary $\sqrt{\text{gʕd}}$. §5 delves into the most complex sentences of this paper, where we have both the auxiliary roots $\sqrt{kwn}$ and $\sqrt{\text{gʕd}}$ with a lexical root, each forming a stem on its own, and each having conjugational affixes. Finally, §6 concludes and presents problems and loose ends. The appendix at the end includes an extensive list of Hasawi Arabic examples of different root and conjugation combinations. It also includes a table of the attested and predicted patterns and another table for the logically possible but unattested, and ruled out patterns.

2 Preparatory States, Target States, and the Aspectotemporal Structure

The goal of this section is to justify the existence of the heads [PREPSTATE] and [TARGETSTATE]. I will show how they are realized as conjugations that only affix to lexical stems. Throughout this section, the auxiliary root $\sqrt{\text{gʕd}}$ “sit” will appear, but I postpone focusing on it (see §4). §2.1 introduces the heads [PREPSTATE] and [TARGETSTATE] and the justification for putting them lower than the auxiliary $\sqrt{\text{gʕd}}$. §2.2 introduces a puzzle of asymmetry between lexical roots such as $\sqrt{nwm}$ “sleep” whose verbal stem can be affixed with the PSC realizing [TARGETSTATE], and other lexical roots such as $\sqrt{ktb}$ “write” whose verbal stem can not be affixed with the PSC realizing [TARGETSTATE]. §2.3 introduces the aspectotemporal structure that elaborates informally on the semantics of [PREPSTATE] and [TARGETSTATE] and explains why we obtain the aforementioned selectional puzzle. §2.4 further illustrates the aspectotemporal structure with more roots in order to test its validity. §2.5 discusses further the distinction between the roots $\sqrt{nwm}$ “sleep” and $\sqrt{ktb}$ “write”. §2.6 notes semantic work by Piñón (1997) that may capture formally the aspectotemporal structure to be introduced. Finally, §2.7 concludes.

2.1 The Target State and Preparatory State

Consider sentences (7a) and (7b), where we have the lexical root $\sqrt{nwm}$ "sleep"[13] and the auxiliary root $\sqrt{\text{gʕd}}$ "sit" in both sentences. I will first elaborate briefly on the semantics of the conjugations affixed to the lexical stems, before I justify their positions relative to the auxiliary root $\sqrt{\text{gʕd}}$ and the conjugation affixed to its stem.

(7) a. χaːləd gaːʕəd naːjɪm
Khalid sit.PSC.SG.M sleep.PSC.SG.M
Obtained: "Khalid is sleeping (for a long while now)."
Unobtained: "Khalid is falling alseep."

b. χaːləd gaːʕəd ji-naːm
Khalid sit.PSC.SG.M 3-sleep.PC.SG.M
Obtained: "Khalid is falling asleep."
Unobtained: "Khalid is sleeping."

Both sentences are very similar: They have the same subject and the exact auxiliary verb. Both are in the present tense too, and in terms of aspect, both have some kind of an "on-going" reading. But there is one crucial difference between them: The conjugation that attaches to the lexical stem. In (7a), the lexical stem is affixed with the PSC realizing the [TARGETSTATE] head, yielding a "sleeping" reading. On the other hand, in (7b), the lexical stem is affixed with the PC realizing the [PREPSTATE] head, yielding a "falling asleep" reading. I will elaborate more on the semantic differences in §2.3. But in a nutshell, the main difference is that the [TARGETSTATE] head denotes a state while the [PREPSTATE] head denotes a process of achieving said state.

As a reminder, I postpone talking about the semantic contribution of the (durative) auxiliary root $\sqrt{\text{gʕd}}$ to §4 (which is related to the "for a long while now" translation in (7a). However, I need to talk about the conjugation on its stem. Specifically, it is important to point out that in (7a), we have two different conjugations, despite both verbal stems appearing in the PSC and thus giving an illusion of PSC doubling. The PSC on the auxiliary stem realizes the head [EPISODIC], which denotes a single state/event as opposed to the [HABITUAL].[14] On the other hand, the PSC on the lexical stem in (7a) is the one that realizes the [TARGETSTATE] head. The evidence for this comes from the obtained readings in (7). If the PSC realizes the head [TARGETSTATE], and if the head [TARGETSTATE] denotes the "sleeping" reading, then we would expect the "sleeping" reading to be available in (7b) if we assume that the PSC affixed to the auxiliary stem is the one that realizes the [TARGETSTATE] head. However, this is not the case. We only obtain a "sleeping" reading in (7) if the lexical stem is affixed with the PSC, and not the auxiliary stem. Sentence (8) illustrates this point further:

(8) χaːləd ja-gʕəd naːjɪm
Khalid 3-sit.PC.SG.M sleep.PSC.SG.M

[13] Due to morphophonology, glides that are part of the underlying root may undergo deletion with the suffixal and prefixal conjugations (/nawam/ stems becoming [naam]), or /w/ > [j] assimilation in the PSC. See Brame (1970) for the morphophonology of Arabic glides.

[14] See §3.2 for elaboration on both heads and §4.3 for further evidence that the two PSCs are different.

Obtained: "Khalid keeps sleeping."
Unobtained: "Khalid keeps falling asleep."

In (8), the lexical stem is affixed with the PSC realizing [TARGETSTATE], while the auxiliary stem is affixed with the PC expressing the head [HABITUAL]. By considering (7) and (8) together, we can clearly see that if there are two verbal stems, an auxiliary stem with the root $\sqrt{\text{gʕd}}$ and a lexical stem, the "target state" reading ("sleeping", with this particular lexical root) will only be available if and only if the PSC is affixed to the lexical stem. By the same logic, we only obtain a "falling asleep" reading if and only if the lexical stem, and not the auxiliary stem, is affixed with the PC realizing the head [PREPSTATE]. This is clearly a constraint on which verbal stem a conjugation can affix to: Whenever there are multiple verbal stems[15] as in (7) or (8), it isn't the case that the conjugations that realize the heads [TARGETSTATE] and [PREPSTATE] can affix to auxiliary stems, and it isn't the case that the conjugations that realize the aspectual heads [EPISODIC] and [HABITUAL] can affix to the lexical stems. This is why I take the heads [TARGETSTATE] and [PREPSTATE] to be below the auxiliary root $\sqrt{\text{gʕd}}$, as they never affix to $\sqrt{\text{gʕd}}$'s stem and instead attach to the lexical stem which is c-commanded by $\sqrt{\text{gʕd}}$. By the same logic, the heads [HABITUAL] and [EPISODIC] are syntactically above the lexical stem and the conjugations attached to it.

It is important to point out that in (7) and (8) the auxiliary verbs did not impose any morphological requirement on the verbs they c-command. Despite having the same auxiliary root surfacing with different conjugations on the auxiliary stems, there are no selectional requirements on the lexical verb. This is unlike languages such as English, where an auxiliary verb such as perfect "have" requires the verb directly after it to be in the past participal (have eaten/have been eating). The conjugations that appear on the lexical stems in (7) and (8) are independent of the auxiliary verb or its elements. To substantiate this claim further, consider the sentences at (9), which are the "simple" (having only lexical verbs) versions of (7):

(9) a. χaːləd naːjɪm
Khalid sleep.PSC.SG.M
Obtained: "Khalid is sleeping."
Unobtained: "Khalid is falling alseep."

b. χaːləd ji-naːm
Khalid 3-sleep.PC.SG.M
Obtained: "Khalid is falling asleep."
Unobtained: "Khalid is sleeping."

In (9), we have no auxiliary verbs, yet the lexical verbs are formally the exact same as their "complex" sentence counterparts in (7): In (7a) and (9a) the lexical stem is affixed with the PSC realizing the head [TARGETSTATE], and thus we have a "sleeping" reading in both. In (7b) and (9b) the lexical stem is affixed with the PC realizing the [PREPSTATE] head and thus we have a "falling asleep" reading in both. So far, in my survey of auxiliary roots such as $\sqrt{\text{gʕd}}$ in my dialect, the auxiliary roots,

[15] I am talking specifically about sentences that potentially contain multiple auxiliary verbs and a lexical verb. Thus I am ignoring coordination of clauses or coordination of lexical verbs.

their stems[16], or their conjugations do not impose any morphological requirement on the verbs that appear after them.

Summarizing, the heads [TARGETSTATE] and [PREPSTATE] are realized by conjugations that affix only to lexical stems. The former denotes a state and thus yields a "sleeping" reading with the root $\sqrt{nwm}$, while the latter denotes a process of achieving the aforementioned state and thus yields a "falling asleep" reading. The presence of an auxiliary verb does not affect the morphology of the verb it c-commands.

Next, I turn to the "Poor Suffixal Puzzle", where there is a semantic constraint on the roots whose stem can be affixed with the PSC realizing the head [TARGETSTATE].

2.2 $\sqrt{nwm}$-Like Roots and $\sqrt{ktb}$-Like Roots

Not all roots behave as $\sqrt{nwm}$ above in (7). Consider the root $\sqrt{ktb}$ "write". Applying this root to a similar pattern yields us (10):

(10) a. * χaːləd gaːʕəd kaːtəb
Khalid sit.PSC.SG.M write.PSC.SG.M
Intended: "Khalid is writing (for a long while now)."

b. χaːləd gaːʕəd ja-ktəb
Khalid sit.PSC.SG.M 3-write.PCSG.M
"Khalid is writing."

In (10a), unlike (7a), we get an ungrammatical sentence: The verbal stem of the root $\sqrt{ktb}$ can not be affixed with the PSC realizing [TARGETSTATE]. On the other hand, it can be affixed with the PC realizing [PREPSTATE], as in (10b). I call this the "Poor Suffixal Puzzle": Why is it that the stem of a root such as $\sqrt{nwm}$ "sleep" can be affixed with the PSC in (7a) and yield an on-going target state reading, but the stem of a root such as $\sqrt{ktb}$ "write" as in (10a) can not?

One might question the innocence of the auxiliary $\sqrt{\text{gʕd}}$ "sit" here, but it really has no bearing on this issue. We have seen above in §2.1 that the conjugations expressing the heads [TARGETSTATE] and [PREPSTATE] are independent of the auxiliary verb. However, just like (9a) above, $\sqrt{ktb}$ can appear in simplex sentences with its verbal stem affixed with a PSC, which is shown in (11b). Here, we face the same dilemma of the root $\sqrt{ktb}$ not obtaining an on-going reading. I provide (11a) in order to compare the roots $\sqrt{nwm}$ and $\sqrt{ktb}$ with respect to the readings that they may yield when both their stems are affixed with a PSC in simple sentences.

(11) a. χaːləd naːjɪm l-ħiːn
Khalid sleep.PSC.SG.M DEF-interval
Obtained: "Khalid has slept now."
Obtained: "Khalid is sleeping now."

[16] The verbal roots in (7) and (8), whether lexical or auxiliary, are associated to the so called Form I. As I have noted before, the auxiliary root $\sqrt{\text{gʕd}}$ always appears with Form I. However, for the lexical root $\sqrt{nwm}$ this is accidental. Other lexical roots take different verbal forms, and they can appear with the same auxiliary verb in (7) or (8). Thus the auxiliary verb does not impose any Form requirements on the lexical root it c-commands.

b. χaːləd kaːtəb l-waːdʒəb l-ħiːn
Khalid write.PSC.SG.M DEF-homework DEF-interval

Obtained: ”Khalid has written the homework now, (why are you scolding him?)”
Unobtained: ”Khalid is writing the homework now.”

In (11a), the stem of the root $\sqrt{nwm}$ is affixed with the PSC. The sentence may yield a perfect (reference time is after event time) reading or an on-going target state “sleeping” reading. On the other hand, in (11b) the stem of the root $\sqrt{ktb}$ is affixed with a PSC, which allows the perfect reading but disallows the on-going reading. Because of this, (11b) is similar to (10a) above in disallowing an on-going reading when the stem of the root $\sqrt{ktb}$ is affixed with a PSC. That’s why whether the auxiliary is present or absent does not change the fact that the combination of the verbal stem of the root $\sqrt{ktb}$ affixed with the PSC does not yield an on-going reading.

The attentive reader might question how the perfect reading arises in these sentences. The answer comes from facts beyond the scope of this paper: The PSC, appearing in sentences without auxiliaries, may yield other semantics such as recent past, remote past, or the experiental perfect (similar to Mandarin’s experiential particle guo4 過). The problem of where do these semantic elements sit in terms of the proposed structure in (1) is beyond the scope of this paper.

Returning to the poor suffixal puzzle, I hypothesize that there is an aspectotemporal difference between $\sqrt{nwm}$ “sleep”-like roots and $\sqrt{ktb}$ “write”-like roots. The next section explains this proposal and elaborates more on the semantics of [PREPSTATE] and [TARGETSTATE].

2.3 The Aspectotemporal Structure

Consider a context where a babysitter is holding a baby. The baby becomes sleepy. Then, he begins to fall asleep. After a lullaby by the babysitter, the baby falls asleep. The baby is now sleeping. The state of the baby being asleep continues for three hours, whereafter the baby wakes up. This context can be illustrated informally with Figure 1:

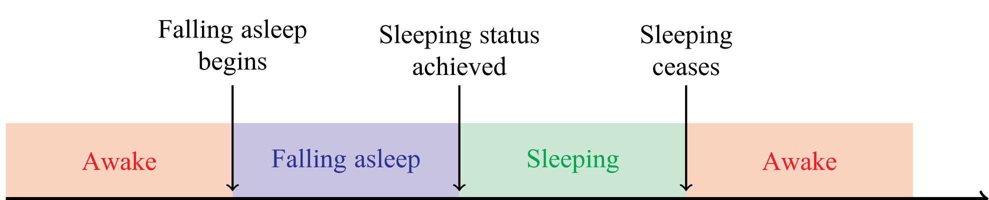


Figure 1: Aspectotemporal Structure of Sleeping

What interests us in Figure 1 are the blue and green zones. Morphologically, the PSC realizes the head [TARGETSTATE] which indicates semantically that the event denoted by the lexical verb is in its target state (the green zone of ”sleeping”). On the other hand, the PC realizes the head [PREPSTATE] which indicates that the event denoted by the lexical verb is in its preparatory state (the blue zone of "falling asleep"). With the above context in mind, consider the sentences in (12):

(12) a. l-walad na:jɪm
DEF-child sleep.PSC.SG.M
"The child is sleeping."

b. l-walad ga:ʕəd ji-na:m
DEF-child sit.PSC.SG.M 3-sleep.PC.SG.M
"The child is falling asleep."

(12a) expresses that the on-going target state obtains after the baby has successfully fallen asleep, which is encoded morphosyntactically by the PSC realizing the [TARGETSTATE] head being affixed to the verbal stem. On the other hand, (12b) expresses the on-going transitory or preparatory state of the baby falling asleep, which is encoded morphosyntactically by the PC realizing the [PREPSTATE] head being affixed to the stem of the lexical root $\sqrt{nwm}$. The morphosyntax differs in interesting ways from English. While both examples share the progressive form in English, in Hasawi Arabic, they differ in that the conjugation on the lexical stem is the deciding factor for whether a target state or a preparatory state reading is obtained. In English, morphemes such as fall may appear in order to yield the preparatory state reading.

With the aspectotemporal structure of Figure 1 in mind, I assume that "sleep"-like roots all can have this structure. In contrast, I am assuming that "write"-like verbs have an aspectotemporal structure as in Figure 2:

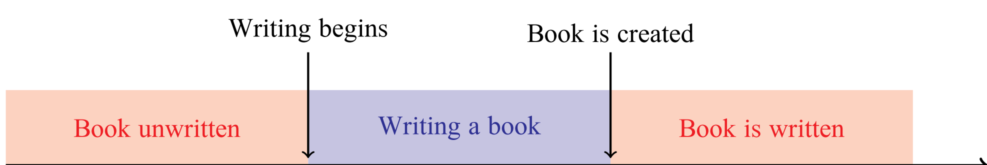


Figure 2: Aspectotemporal Structure of Writing a Book

Thus I am assuming that "write"-like roots do not express a target state. In the context of writing a book, there is only the preparatory state of writing it and it being written. There is no intermediate step in between, unlike sleeping where one can start falling asleep, enters the state of being asleep, and then wakes up. This intermediate step represented above in Figure 1 with the green zone is the crucial difference between write and sleep (though see below in §2.5 for a "coercion" of "write"-like roots). Thus this explains why sentences such as (10a) and (11b) do not yield an on-going writing reading: Since these lexical roots lack the target state stage, the PSC can not express a target state reading with them. While this discussion of semantics is very preliminary and informal, see §2.4 for a discussion on a possible lead to this structure in terms of Piñón (1997)'s event boundary semantics. In the next section, I look at other roots to test the above aspectotemporal structures.

2.4 More Aspectotemporal Structure Demonstrations

So far, I have been mentioning "sleep-like" roots without deviating from "sleep". This is a good place to talk about other roots and see how they fare with our aspectotemporal structure in Figure 1. Some of the roots that behave such as "sleep" and may express a preparatory state or a target state reading include $\sqrt{\text{gʕd}}$ "to sit (down)", $\sqrt{wgf}$ "to stand (up)", $\sqrt{\text{fs}^{\text{ʕ}}\chi}$ "to take off (clothes/equipment)", and $\sqrt{lbs}$ "to put on/wear (clothes/equipment)", among many others. The last root, $\sqrt{lbs}$, is a very good root to show the lexical and functional semantic differences between HA and English. Consider a new context, in a similar vein to the babysitter one above, where I will give examples about $\sqrt{lbs}$ "to put on/wear" and its counterpart $\sqrt{\text{fs}^{\text{ʕ}}\chi}$ "to take off":

Context: Abdullah works in a nuclear reactor. In order to enter a specific room, he has to put on his hazmat suit. A TV crew sends in a robot in order to film Abdullah's work for a documentary of some sort. The robot films Abdullah's journey beginning with his putting on the hazmat gear, walking around the radiated room, and then finally, stripping down the equipment:

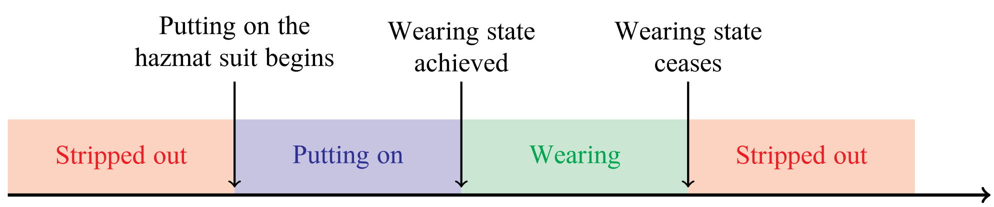


Figure 3: Aspectotemporal Structure of Putting on a Hazmat Suit

With the above context in place, consider sentence (13), uttered when the crew is filming Abdullah before his entering of the room:

(13) ʕabdalːah gaːʕəd ji-lbas ziː l-hazmaːt
Abdullah sit.PSC.SG.M 3-wear.PC.SG.M suit DEF-hazmat
"Abdullah is putting on the hazmat suit."

While Abdullah is inside the radioactive room, a crew member might remark that "Abdullah is protected because he…."

(14) laːbəs ziː l-hazmaːt (ʔal-ħiːn)
wear.PSC.SG.M suit DEF-hazmat (DEF-interval)
"...is wearing the hazmat suit (right now)."

(13) and (14) show a very similar pattern to the examples of the root $\sqrt{nwm}$ in (12). Not only that, they also share the same aspectual semantics: In both (12a) and (14), the lexical stem is affixed with the PSC, and there is an on-going target state reading of sleeping or wearing a suit. In both (12b) and (13), the lexical stem is affixed with the PC, and in both sentences there is an on-going preparatory state of falling asleep or putting on a suit. Notice how we were able to obtain the same preparatory state

and target state readings, despite the roots differing in lexical semantics. “Sleep” is usually taken to be a stative and an unergative predicate (barring causation like “put to sleep”). On the other hand, “Put on X” is an “eventive/process” and an ergative predicate. Thus it seems that we are on the right track if we assume the aspectotemporal structure to be the common thing between these roots.

Back to the hazmat suit context, I will finish it off with another example. Imagine that Abdullah is finally done with his shift inside the radioactive room. The next step is to take off the hazmat suit. Thus a crew member might say something like (15) when Abdullah is getting ready to get out of the radioactive room:

(15)	ʕabdalːah	gaːʕəd	ji-fsˤaχ	ziː	l-hazmaːt
	Abdullah	sit.PSC.SG.M	3-take.off.PC.SG.M	suit	DEF-hazmat
	“Abdullah is taking off the hazmat suit.”				

(15) is the counterpart to (13): Both are a preparatory stage of putting on and taking off the hazmat suit, respectively. The target state counterpart to (14) would be (16), where Abdullah is no longer wearing the hazmat suit:

(16)	ʕabdalːah	faːsˤəχ	ziː	l-hazmaːt	(l-ħiːn)
	Abdullah	take.off-PSC.SG.M	suit	DEF-hazmat	(DEF-interval)
	“Abdullah has taken off/is stripped out of the hazmat suit (now).”				

To my knowledge, the best way to translate (16) is with an English perfect structure “has taken off”, unlike the target state of “is sleeping” and “is wearing”, there is no progressive structure (which might be a lexical gap).

Finally, consider the root √ʃjl “to lift/hold up”[17]. (17) and (18) show the preparatory and target state sentences in HA respectively. Note that (17) is said in a context of one lifting up process, not an iterative one:

(17)	ʕabdalːah	gaːʕəd	ji-ʃiːl	barbell
	Abdullah	sit-PSC.SG.M	3-lift.up.PC.SG.M	barbell
	“Abdullah is lifting up a barbell.”			

(18)	ʕabdalːah	ʃaːjɪl	barbell
	Abdullah	lift.up-PSC.SG.M	barbell
	“Abdullah is holding up a barbell.”		

What interests us with this root, is the fact that we still get the morphosyntactic structure in (18) with target state semantics, despite the state of holding something up requiring agentivity: A person who is holding up a barbell has to keep or exerts the agentivity/willingness of holding something up. This is unlike predicates such as “being asleep” or “wearing a T-shirt”, where one does not need to exert agentivity in those target states[18]. With this, we can say that there is no variable of (non-)agentivity at play here: The natural class that these roots have in common is their ability to

[17] I have conflicting reports about the usage of these two predicates. Some find ”is lifting up” permissible in both target state and preparatory state readings, others find it only permissible in preparatory state readings.

[18] I am indebted to Ruoying Zhao for pointing this out to me.

express a target state which can be reversed, as in someone putting on a T-shirt, then wearing it, and then stripping it off, to then maybe begin another cycle of putting it on again. And just like $\sqrt{nwm}$ "sleep", figure 4 gives the aspectotemporal structure for $\sqrt{\text{ʃjl}}$ "to lift/hold up":

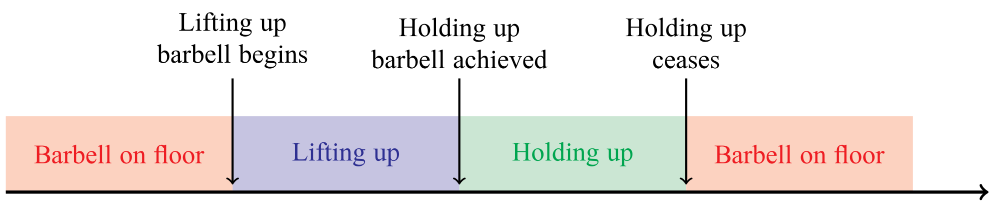


Figure 4: Aspectotemporal Structure of Lifting up a Barbell and Putting It down

In conclusion, variables such as argument structure (ergatives vs unergatives), Vendler classes (state predicates versus event predicates[19]), and agentivity, are not factors on the morphosyntax on whether a root can have a target state reading: What matters is the aspectotemporal structure.

The proposal so far is very informal, semantics wise. I have no formal semantic account that captures the aspectotemporal structure. However, I mention briefly potential formal semantic leads to the aspectotemporal structure in §2.6, after I have a final word on the "write" vs "sleep" distinction.

2.5 $\sqrt{nwm}$-Like Roots Versus $\sqrt{ktb}$-Like Roots: A Superficial Stipulation?

In Figure 2, I have claimed that "write"-like verbs may not express a target state, and that therefore, the PSC does not receive a target state interpretation. However, upon closer scrutiny, they can. This problem is not obvious, but it is what Klein (1994) has noted about "source state" (his preparatory state) and target state distinction. Predicates such as "to write" and "to die" have no further distinction: "Once dead, he is forever dead." Klein (1994, P23). Similarly, once a book is written, it is always written. Thus it is odd to say:

(19) # John is still dead.

(20) # John has still written the book[20].

However, if there is a different context, then these predicates could indeed express a target state. Take for example the verb "to die". In our real world knowledge, such a verb would express an aspectotemporal structure like write's Figure 2. Imagine a situation where a person is in the hospital due to some grave illness. Then, saying the following is appropriate:

[19] As far as Vendler classes go, Vendler (1957) was concerned with "verbs" in English, but here we are concerned with roots. It seems that HA is a language that "builds up" the classes morphosyntactically, as one begins with the root and then keeps adding morphological material until the derivation stops. If this is true, then HA supports decompositional approaches to inner/lexical aspect.

[20] I am ignoring here the reading of "still", paraphrasable as "despite…".

(21) John is dying (from his grave illness).

Figure 5 is a copy of Figure 2:

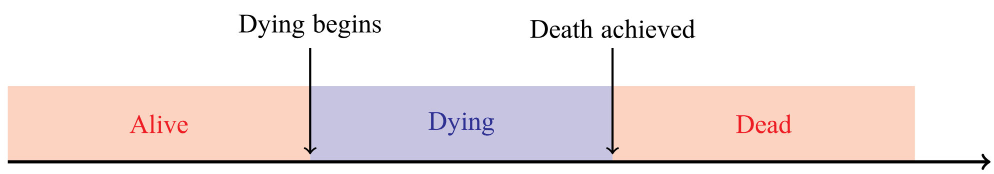


Figure 5: Aspectotemporal Structure of John Dying

In this real world context, (19) is odd. However, if one were to journey to the whimsical world of video games and their logic (especially ones that allow resurrection), then the temporal structure would be different:

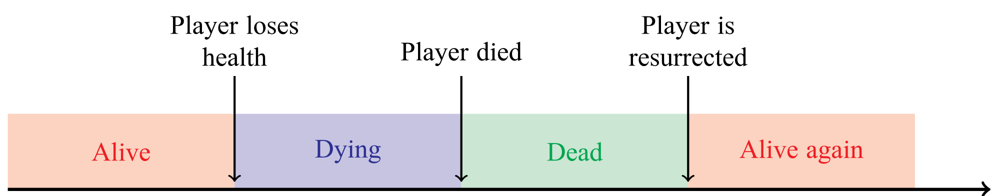


Figure 6: Aspectotemporal Structure of a Player Dying and Resurrecting

Given a context like Figure 6, (19) would be more than appropriate not only for English, but also for HA. Consider the following HA sentences with the root $\sqrt{mwt}$ "to die, death, dead":

(22) John gaːʕəd ji-muːt !
John sit-PSC.SG.M 3-die.PC.SG.M
"John is dying! (His health is dropping!)"

(23) John l-al-ħiːn majːɪt ! gaʕd-uː-h !
John DAT-DEF-interval dead-PSC.SG.M ! resurect.IMP-PL-him !
"John is still dead! Ressurect him!"

In (22), once again we have a preparatory state (dying is on-going), as encoded by the PC on the lexical verb realizing the head [PREPSTATE]. In (22) and (23), the lexical stem is affixed with the PSC realizing the [TARGETSTATE].

For the verb write, one needs to be further imaginative in order to force a target state reading. Consider the following, very contrived context:

Context: UCL's website is bugged. There is a bug that for some reason allows only one language per day to show on the website. The languages on display rotate from English to Japanese to Russian and back to English. Yesterday, it was Japanese. Today, Abdullah opens his computer, and counter to his expectation, the website is still written in Japanese, so he remarks and says: "That's weird...."

(24) la-l-ħiːn kaːtb-iːn l-mawqəʕ b-əl-jaːbaːni
DAT-DEF-interval write.PSC-PL DEF-website INS-DEF-Japanese
"They still have written the website in Japanese."

I am unsure of how to translate this sentence. What seems to be happening here is a coercion: Texts usually stay written. However, a website might change its text and not only that, might rotate with another language, as is the case here. It seems that this language rotation is what brings out the target state reading. Thus the problem could be world knowledge, instead of inherent lexical semantics of these predicates. If so, then there is no actual distinction between "sleep"-type verbs and "write"-type verbs. Regardless, unless convoluted contexts such as video game logic or bugged websites are encountered, I will stick to Klein's "Once dead, always dead" distinction (once dead/written, there is no further "undead" or "unwrite" state to the right of the aspectotemporal axis).

2.6 Adopting a Semantics

A formal way of capturing the aspectotemporal structure would be Piñón Piñón (1997)'s proposal about the ontology of event semantics, where not only do eventualities (states, processes, and events, which he calls "happenings") are taken for granted to be entities, but also "boundary happenings": Entities which have the temporal trace of points in the time-axis and signify the boundaries of happenings. He illustrates with the following figure (adopted from Piñón (1997, P11)):

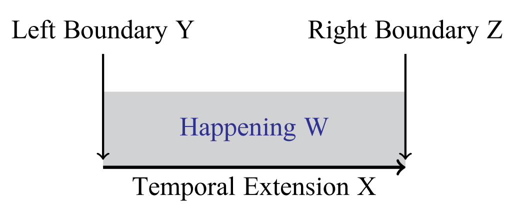


Figure 7: A Structure of an Eventuality

Figure 7 represents a happening W. This happening consists of a temporal extension X and two boundary happenings, Y and Z. Thus instead of relying on the current informal aspectotemporal structure above, one could adopt Piñón's ontology in order to give precise semantics to the heads [PREPSTATE] and [TARGETSTATE]. For [PREPSTATE], it could take the verbal predicate and give it a temporal extension, as well as a visible right boundary that it attempts to cross. On the other hand, the head [TARGETSTATE] might take a verbal predicate and give it the structure of a happening without an event of crossing its right boundary. If these hunches are on the right track, then this would be a nice semantic supplement to the morphosyntactic structure of (1). However, I will leave this implementation to future work.

2.7 Summary

Concluding §2, I have introduced the heads [TARGETSTATE] and [PREPSTATE] which are expressed as conjugations affixed to lexical stems in order to account for the target state ”sleeping” and preparatory state ”falling asleep” readings. Evidence from ”complex” sentences not only suggest that these heads attach only to lexical verbs, but that they are also lower than auxiliary stems and the conjugations that affix to them, such as [HABITUAL] and [EPISODIC]. I have also shown that conjugations act independently from each other, as auxiliaries do not impose any morphological requirements on the lexical verbs. The [TARGETSTATE] and [PREPSTATE] heads were used also as an explanation for the poor suffixal puzzle, as they express the aspectotemporal structure in Figure 1 and Figure 2. The conclusion was that some roots such as ”sleep” were able to express both target and preparatory states, in contrast to roots such as ”write” which only express preparatory states (coercion aside). Other factors such as argument structure, Vendler classes, and agentivity did not play a role in clarifying the natural classes of these roots. Morphosyntactically, HA, in comparison to English, has no different roots or lexical verbs to distinguish the preparatory state (fall asleep) from the target state (sleep): HA uses conjugations in order to mark the distinction, whereas English uses various morphemes such as prepositions (sit down, stand up, etc) or other “fall” like elements, or even suppletion (put on/wear). In the next section, I will take a step back from auxiliaries and show how all of the conjugations expressing various tense and aspect heads can affix into verbal stems in simple verb sentences.

3 Simple Sentences

In this section, I will go through each of the heads expressed by conjugations in (1). The general idea is that, just like in (6) above, there will be a single lexical stem which undergoes head-movement into a higher head that is realized by a conjugation. By doing this, I will be able to highlight the semantic contribution of each head independently and reinforce the statement that the conjugations are independent of other conjugations and auxiliaries as we only have two variables here: The lexical stem and a conjugation. In §3.1, I will focus on simple sentences with PC expressing [PREPSTATE], as we have already seen simple sentences with PSC expressing [TARGETSTATE]. §3.2 focuses on the PSC expressing [EPISODIC] and PC expressing [HABITUAL]. §3.3 focuses on the SC expressing [PAST], the PC expressing [FUTURE], and the (morphophonologically) problematic [PRESENT]. §3.4 discusses the multiple readings available with the [HABITUAL], [PAST], and [FUTURE]. §3.5 elaborates further on tense and shows how ”be-support” is invoked in order to support the realization of multiple conjugations. §3.6 concludes.

3.1 Simple Preparatory State

§2 already talked in-depth about the states [PREPSTATE] and [TARGETSTATE]. I have given abundant examples with a single verbal stem that is affixed with the

PSC expressing [TARGETSTATE]. However, all of the examples that included a PC that realizes the head [PREPSTATE] included the auxiliary $\sqrt{\text{gʕd}}$. This is because the context that yields a simple [PREPSTATE] sentence is hard to come by. It needs to be an event which is on the verge of its right boundary. Consider the following context:

Context: Abdullah picks up a leaflet. While he is reading it, Khalid asks him “What are you doing?” To which Abdullah, while reading up the last couple of words of the leaflet, replies with:

(25) ʔa-gra waraq-ah
1.SG-read.PC paper-SG.F
“I am reading a leaflet.”

In this context, where Abdullah is on the verge/has just ended reading the leaflet, one can produce simple [PREPSTATE] sentences. I can not use the auxiliary $\sqrt{\text{gʕd}}$ “sit” here:

(26) # gaːʕəd ʔa-graː waraq-ah
sit-PSC.SG.M 1.SG-read.PC paper-SG.F
“I am reading a leaflet.”

Unless Abdullah keeps reading the leaflet, (26) is infelicitous. Note that I am using an example where the verbal agreement has first person, as this facilitates obtaining the context further. This is because it is hard to know whether someone is on the verge of reading up a leaflet or not. As such, it appears that the variable of evidentiality is also important here. If I were to use a third person example, the speaker then has to somehow know that Abdullah is reading his last words.

Questions of right boundary verge and evidentiality aside, one might question if the important variable in (25) is the fact that it is a reply, and thus the auxiliary verb is required to be deleted in replies in HA. This is not the case. Consider (27):

Context: Khalid asks Abdullah “What are you doing?” while Abdullah is in the middle of reading Sun Tzu’s “The Art of War”:

(27) ((ʔaːna) gaːʕəd) ʔa-gra Sun Tzu
((I) sit.PSC.SG.M) 1.SG-read.PC S T
”I am reading Sun Tzu.”

The reply in (27) has three options: (i) a full sentence that includes the subject and the auxiliary. (ii) elision of the subject only. (iii) elision of both the subject and the auxiliary[21]. I take this to show that being a reply is not the factor making (26) infelicitous, but the “very short” or “on the verge of ending” context that demands a simple [PREPSTATE] sentence.

To confirm further this very unique context and its requirement of a PC on the lexical stem, consider again the context in (25), this time using different conjugations in (28). Affixing either a PSC or a SC on the lexical stem yields an infelicitous sentence:

(28) #gaːɹi / #gəɹeː-t waraq-ah
read.PSC.SG.M / read.SC-1.SG paper-SG.F
Intended: I am reading a leaflet.”

[21] Eliding the auxiliary without eliding the subject is ungrammatical to me.

Based on the above facts, I take that (25) is a non-durative, on the verge of ending context that requires a PC on the lexical stem, and that this PC expresses the head [PREPSTATE][22]. The infelicity of (26) can be explained as the auxiliary √gʕd "sit" introducing durative semantics (see §4 for more elaboration on √gʕd's semantics). However, further insights from the literature on evidentiality and from languages that show non-durative morphology are required in order to corroborate the claims made here.

3.2 The Aspectual Heads Habitual and Episodic

The head [HABITUAL] is realized by the PC. It is appropriately translated with the habitual meaning of English's simple present. (29) is a sentence that expresses a habit (daily routine) with the root √ʃjl "to lift up/hold up":

(29) χaːləd ji-ʃiːl ʔaθgaːl (kɪl joːm)
Khalid 3-lift.PC.SG.M weight.PL (every day)
"Khalid lifts up/holds up weights (every day)."

The head [HABITUAL] basically expresses generic statements/dispositions/contextual habits of individuals. There is some discussion to be had with the translation being a "lifts up" or a "holds up" reading, for which see §3.4. Having said that, we will move into the [EPISODIC] which is a bit problematic.

The head [EPISODIC] is the reverse of the head [HABITUAL]: Instead of an event/state recurring (at a certain frequency), the [EPISODIC] denotes a single occurrence of an event/state. This head in simplex cases only occurs with a handful of predicates, mainly stative ones such as hate and love:

(30) χaːləd ji-ħəbː s-salamon, bas li-ʔanː ʔabdalːah
Khalid 3-love.PC.SG.M DEF-salmon, but DAT-that Abdullah
sawːa-a-h kaːɹh-ah / jɪ-kɹah-ah
make.SC-3.SG.M-it hate.PSC.SG.M-it / 3-hate.PC.SG.M-it
"Khalid loves salmon, but because Abdullah made it he is hating it."

In (30) there are three verbal roots. The first is √ħb "love" in the first clause, √swj "do/make" in the "because" clause, and √kɹh "hate" in the second clause. √ħb's verbal stem is affixed with the PC expressing [HABITUAL], giving a habitual state reading "Khalid loves salmon". This habit is temporarily halted due to a certain context (Abdullah's cooking it). This is evidenced by the verbal stem of the root √kɹh "hate" being affixed with the PSC which I take to express the head [EPISODIC]. The sentence becomes infelicitious if the stem of "hate" is affixed with the PC, which sounds like a contradiction to my ears. As for English, it is known in the aspectual literature that the progressive coerces stative predicates into a temporary reading. It seems in (30) that both the Hasawi PSC and the English progressive are morphosyntactic reflexes of these temporary states of loving or hating.

[22] Another variable that might be considered is tense "distance", as in, this could be an immediate past reading. Further tests and consultation of the semantics of the tense distance literature are warranted.

However, problematically, I have no explanation for why this temporary reading does not obtain with other roots such as $\sqrt{nwm}$ "sleep" or $\sqrt{ktb}$ "write". One could see the head [TARGETSTATE] being inherently episodic/temporary (there is a state of being asleep that is not quantified over by a habitual operator), and thus the possibility of the [EPISODIC] head being redundant surfaces. However, the usefulness of postulating the head [EPISODIC] will become clear in §4.3, where we see how the conjugation expressing it and the conjugation expressing the head [HABITUAL] may affix on the stem of the auxiliary root $\sqrt{\text{gʕd}}$ "sit" in order to derive single event readings and habitual event readings respectively. There is also the fact that two PSCs can appear in the same sentence when two verbal stems (auxiliary and lexical) are present. I will also present work based on Bedouin Palestinian Arabic where similar facts hold, namely, that there is a correlation between the appearance of the PSC on the auxiliary stem and episodic (non-habitual) semantics.

3.3 Tense

In this section, I will elaborate on the tense heads [PAST], [FUTURE], and then [PRESENT]. The heads [PAST] and [FUTURE] have clear morphological reflexes. However, things are not so clear with the [PRESENT]. I will begin with the [PAST] and [FUTURE] heads, before moving on to the problematic [PRESENT], which I take to be phonologically null. Further tense illustrations with "be-support" are introduced in §3.5.

Beginning with the head [PAST], consider sentence (31) which has the root $\sqrt{nwm}$ "sleep". Its stem is affixed with a SC:

(31) χaːləd naːm (ʔams / lħiin / *bukɹah / *baʕd ʔɪʃwæj)
Khalid sleep-SC.3.SG.M (yesterday / now / *tomorrow / *after little)
"Khalid fell asleep/slept (yesterday / now / *tomorrow / *soon)."

(31) expresses a past tense reading. The verb is also compatible with adverbials that situates the reference time prior to utterance time, such as "yesterday". It is ungrammatical with adverbs that situates the reference time posterior to utterance time, such as "tomorrow" or "soon". (31) can also yield either a past preparatory state reading "fell asleep" or a past target state "slept" reading (just like the habitual in (29). I will return to this point in §3.4). Based on these facts, I take the suffixal conjugation to be the realization of the tense head [PAST].

The head [PAST] is easy to explain. On the other hand, the head [FUTURE] needs more elaboration. Consider sentence (32) which is the future counterpart to (31):

(32) χaːləd b-iː-naːm (*ʔams / lħiin / bukɹah / baʕd ʔɪʃwæj)
Khalid FUT-3-sleep.PC.SG.M (*yesterday / now / tomorrow / after little)
"Khalid will fall asleep/sleep (*yesterday / now / tomorrow / soon)."

In (32), the verb contains two prefixes, one is the agreement prefix /ji-/ that occurs with the PC, and prefixed to it is the future morpheme /b-/. In sharp contrast to the past sentence in (31), (32) is compatible with adverbs that situate the reference time to the posterior of utterance time, and is ungrammatical with adverbs such as "yesterday."

However, both (31) and (32) are similar in being compatible with the adverb ”now”, which yields an immediate past/future reading.

Having established claims that (32) is a future sentence, I turn to the prefix /b-/. One might ask the question: “Why not postulate the future prefix /b-/ independently from the PC?”. The problem is that, just like how there is a one-to-one correspondence between the environment of the PC and the agreement affixes that appear in it (such as the prefix /ji-/), there is also a one-to-one correspondence between the future prefix /b-/ and the PC: Whenever the future prefix /b-/ affixes to a base, that base is always a verbal stem that is affixed with the PC. Thus if one takes (32) and replaces the PC with either the PSC or the SC, the result is an ungrammatical sentence:

(33) a. * χaːləd b-naːjɪm
Khalid FUT-sleep.PSC.SG.M
Intended: “Khalid will fall asleep/sleep.”

b. * χaːləd b-naːm
Khalid FUT-sleep-SC.3.SG.M
Intended: “Khalid will fall asleep/sleep.”

This is an inherited property from Classical Arabic. In Classical Arabic, the morpheme realizing the future can be a prefix /sa-/ or a free-standing particle /sawfa/. Both morphemes select a verbal stem that is affixed with the PC:

(34) sa-ja-quːl
FUT-3-say.PC ....
“They will say ....” (Quran: Al-baqarah, verse 142)

(35) .... sawfa ju-ʔtiː-hum ʔud͡ʒuːɹa-hum ....
.... FUT 3-give.PC.SG.M-them reward.PL-their ....
“He will give them their rewards....” (Quran: Al-Nissa, verse 152)

While I do not have a peripherastic future morpheme as in (35) in my HA, my future /b-/ prefix works similarly: It yields future semantics (reference time is after utterance time) and imposes conjugational restrictions on the verb it affixes to.

The facts above make it clear that a verb such as biːnaːm “He will fall asleep/sleep” is composed of a root $\sqrt{nwm}$ “sleep”, the stem of Form I which corresponds to the CV-skeleton /CCVC/[23], a prefixal agreement morpheme /ji-/ that marks the conjugational class of the PC, and a prefixal future morpheme /b-/. The root (lexical or auxiliary) and the Form may differ, but the PC and the prefixal future morpheme form a natural class in appearing at the same environment (the future /b-/ selects the PC which in turn selects the set of agreement morphemes). I will simplify this natural class by using the head [FUTURE] while noting the facts above.[24]

[23]Once again, glide morphophonology strikes to make it appear as surface [CVVC].

[24]Alternatively, it could be the case that the PC is the “default” conjugation that appears on the verbal stem, as is argued by authors such as Benmamoun (1999). The prefix /b-/ then attaches to this default conjugation, and thus there is no real selectional restriction by the future prefix /b-/. However, if the prefix /b-/ does not select the PC, then its absence from the poor suffixal and the suffixal will only be accidental.

While the exponents of the past and future tenses can be easily pointed out by the morphological facts above, things are not so clear with the present tense. At least for HA and Classical Arabic, there doesn't seem to be any trace for a present tense morpheme. I will take the stance that there is a phonologically null [PRESENT] head in Hasawi Arabic (and possibiliy by extension, Classical Arabic). I will justify my reasoning for taking this stance below.

There are two important generalizations that lead me to claim a null [PRESENT] head. The first is the fact that, it is a necessary, but not a sufficient condition, for a sentence to contain no [PAST] or [FUTURE] morphology in order for a present tense reading to be available[25]. Second, if the necessary condition obtains (the sentence doesn't contain either the suffixal conjugation realizing [PAST] or the prefix /b-/ realizing [FUTURE]), it isn't the case that the sentence is tenseless, as there are always restrictions on the reference time in any clause. I illustrate the above two generalizations with three types of examples: (i) copular sentences, (ii) target state sentences, and (iii) habitual eventive sentences.

First, consider the copular sentences below:

(36) a. l-mɑːj maːləħ
DEF-water salty
"The water is salty."

b. l-mɑːj ji-kuːn maːləħ fii h-al-wagt
DEF-water 3-be.PC.SG.M salty at DEM-DEF-time
"The water is salty at this time (of the year)"

c. l-mɑːj kaːn maːləħ
DEF-water be-SC.3.SG.M salty
"The water was salty."

d. l-mɑːj b-iː-kuːn maːləħ
DEF-water FUT-3-be.PC.SG.M salty
"The water will be salty."

In (36), there are four copular sentences, each denoting a specific tense. (36a) and (36b) entail only present readings, (36c) a past reading, and (36d) a future reading. The goal of presenting the sentences (36a) and (36b) is to show that the presence or absence of the copula plays no role on whether a present tense reading could be available. What matters is whether there is past or future tense morphology or not. One difference between (36a) and (36b) is that (36b) can be said in contexts where there is modification, either by phrases such as (at this time of year) or by adverbial clauses such as "whenever" or "each time", which appear to license the appearance of the copula. However, the copula's stem is affixed with the prefixal conjugation and no apparent tense morphology.

The facts in (36) are not restricted to HA. Eid (1983, P197-198) has the same pattern in her Egyptian Arabic (EA). (37) is EA's replica of (36)(I have taken the liberty to adjust her glossing a bit).

[25] I am simplifying in this paper, as I am ignoring multiple tense interactions such as main and subordinate clause tenses or phenomena such as sequence of tense.

(37) a. l-mudaɹ:is lati:f
DEF-teacher nice
"The teacher is nice."

b. l-mudaɹ:is ʕadatan b-i:-ku:n lati:f lam:a ʔa-kal:im-u
DEF-teacher usually PROG-3-be.PC.SG.M nice when 1.SG-talk.PC-him
"The teacher is usually nice when I talk to him."

c. l-mudaɹ:is ka:n lati:f
DEF-teacher BE-SC.3.SG.M nice
"The teacher was nice."

d. l-mudaɹ:is ħa-j-ku:n lati:f
DEF-teacher FUT-3-BE.PC.SG.M nice
"The teacher will be nice."

Once again, the copular sentences that obtain a present tense reading are those without a past or future tense morphology, namely (37a) and (37b) . EA is similar to HA, and thus (37c) has a SC on the copula expressing past tense, and (37d) has the prefix /ħ-/ expressing future tense[26][27].

Next, let us return briefly to the head [TARGETSTATE]. The important point with a simple sentence that has the [TARGETSTATE] is that only a present tense reading is available:

(38) χa:ləd wa:gəf
Khalid stand.PSC.SG.M

Obtained: "Khalid is standing"
Unobtained: "Khalid was standing."
Unobtained: "Khalid will be standing."

In (38), the sentence has the lexical stem attached with the PSC expressing [TARGETSTATE], and only a present tense reading is available. Just like the copular sentences above, the absence of the morphology of the head [PAST] and [FUTURE] correlates with the presence of the present tense reading.

Finally, I move to an apparently problematic case, which is the reason for the "but not a sufficient condition" statement above. Consider (39) , which has a lexical stem affixed with the PC. The PC is compatible with the adverbial phrases "every day" and "tomorrow", but it is ungrammatical with "yesterday":

(39) χa:ləd ji-lʕab kuɹah (kɪl jo:m / bukɹah / *ʔams)
Khalid 3-play.PC.SG.M football (every day / tomorrow / *yesterday)
"Khalid plays football (every day / tomorrow / *yesterday)."

[26]EA has the prefix /b-/, but unlike HA it is not the future marker, as that would be the prefix /ħ-/. Although Eid (1983) glosses the the prefix /b-/ as "present", Mughazy (2004) glosses it as PROG. Other Egyptian speakears I have talked to consider it an aspectual prefix that is related to on-going readings or "progressitivity" in general. Regardless of its real status, it isn't a past or a future morpheme, and the reading with it is a present tense reading.

[27]Note how the future prefix /ħa-/ in EA selects the prefixal conjugation, just like HA's future prefix /b-/ or CA's future prefix /sa-/.

In (39) two readings are available, a habitual and a futurate reading. The futurate reading is similar to the one available to English's simple present form, where the event is planned/scheduled to happen in the future. Because of this, there is the possibility for a non-past tense head being present in (39) . However, there is evidence to claim that (39) is ambiguous between a present habitual and a futurate reading. Consider the following context:

Context: Khalid plays football every week. On the other hand, his friend, Abdullah, hates football and doesn't play it. However, he lost a bet to Khalid, and now he must play for a day. Thus Abdullah plans on playing football tomorrow, and then never touching it again.

Based on this context, the following two sentences are true:

(40) χaːləd ji-lʕab kuɹah kɪl ʔəsbuuʕ
Khalid 3-play.PC.SG.M football every week
"Khalid plays football every week."

(41) ʕabdalːah ji-lʕab kuɹah bukɹah
Abdullah 3-play.PC.SG.M football tomorrow
"Abdullah plays football tomorrow."

Both the habitual (40) and the futurate (41) have the lexical stem affixed with the PC. If this is an instance of underspecification, then conjoining the subjects without the adverbs would yield a felicitious sentence, which is not what happens:

(42) # χaːləd wə ʕabdalːah ji-lʕab-uun kuɹah
Khalid and Abdullah 3-play.PC-PL football
Intended: "Khalid and Abdullah play football."

The fact that (42) is infelicitous makes it clear that we have an ambiguous sentence with (39). I assume that the habitual reading in (39) is obtained via the head [HABITUAL] which is realized by the PC, as discussed above in §3.2. As for the availability of the futurate reading, it could be the case that there is another head, say, [FUTURATE], which might introduce the "plan" as an entity, as advocated by Copley (2014). Further elaborations on the futurate is beyond the scope of this paper.

Based off the facts above, I contend that whenever the morphology of the [PAST] and [FUTURE] heads (and whatever realizes the futurate reading) is absent, there is a phonologically null present tense head [PRESENT] which restricts the reference time to the present (utterance time). This captures two facts: (i) sentences without any tense morphology are not free to be located at any reference time (as the unobtained readings show in (38), and (ii) there is a correlation between the absence of overt [PAST] and [FUTURE] tense morphology and the availability of present tense readings. This accounts nicely for null-copula sentences, and simple sentences that have only one verbal stem which is affixed with the PSC realizing [TARGETSTATE] or with the PC realizing [HABITUAL].

3.4 The Availability of Preparatory State and Target State Readings

In this section, I address the problem of multiple readings being available with the habitual in (32), the past in 31, and the future in (32). In these three sentences, one can obtain a preparatory state (falls asleep, fell asleep, will fall asleep) or a target state (sleeps, slept, will sleep) reading. I will use the SC realizing [PAST] as an example, but the logic generalizes to the cases of [HABITUAL] and [FUTURE] too.

The root $\sqrt{nwm}$ "sleep" with its stem affixed with the SC can be used in a context where a patient slept the amount of sleep needed or just fell asleep. In both cases, it is fine to follow both contexts with a clause that either asserts that the sleeping is still on-going, or that it has stopped:

(43) l-maɹiːdˤ naːm χams saʕaːat (wə la-l-ħiːn
DEF-patient sleep-SC.3.SG.M five hours (and DAT-DEF-interval
naːjɪm / wə baʕdeːn gɪʔad)
sleep.PSC.SG.M / and then wake.up-SC.3.SG.M)
"The patient slept for five fours (and is still sleeping / and then woke up).

(44) l-maɹiːdˤ naːm (wə la-l-ħiːn naːjɪm / bas
DEF-patient sleep-SC.3.SG.M (and DAT-DEF-interval sleep.PSC.SG.M / but
gɪʔad bsəɹʕah)
wake.up-SC.3.SG.M quickly)
"The patient fell asleep (and is still sleeping / but woke up quickly).

I have no conclusive evidence for claiming the verb naːm as either underspecified or ambiguous between the target state (43) and preparatory state (44). I will take the absence of a PSC realizing [TARGETSTATE] or a PC realizing [PREPSTATE] to indicate that they are optional, unlike tense which seems to be obligatory in every main clause. I will give examples where they co-occur with the SC realizing [PAST] in the next section.

3.5 Tense and Last Resort $\sqrt{kwn}$ "Be"

In §3.3, I have elaborated on the morphology of tense in simple sentences. However, when two conjugations need to be realized, it seems that a last resort operation of "be-support" activates. Consider the past sentences (45) and (46) in comparison to (43) and (44) above.

(45) χaːləd kaːn ji-naːm
Khalid be-SC.3.SG.M 3-sleep.PC.SG.M
"Khalid was falling asleep."

(46) χaːləd kaːn naːjɪm
Khalid be-SC.3.SG.M sleep.PSC.SG.M
"Khalid was sleeping."

In (45) and (46) , the lexical stems are affixed with the PC realizing the head [PREPSTATE] and the PSC realizing the head [TARGETSTATE], respectively. However, in both sentences, the SC realizing the tense head [PAST] is affixed on the auxiliary stem of the root $\sqrt{kwn}$ "be". This is unlike the sentences in (43) and (44) , where no PSC or PC exist, leading the SC to affix to the lexical stem and thus there is no need for "be-support".

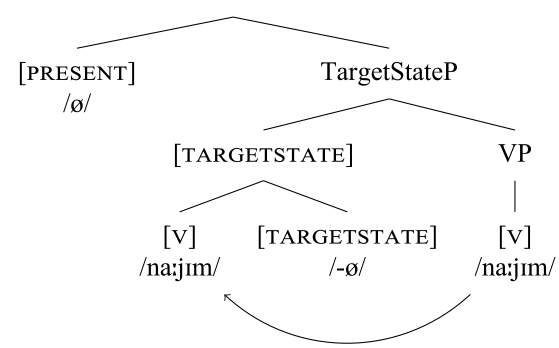
Similar to the complex sentenes in §2.1, we have constraints on which verbal stems conjugations can affix to. If we swap the conjugations on the verbal stems in (45) and (46) , the result is ungrammatical sentences:

(47) * χa:ləd ji-ku:n na:m
Khalid 3-be.PC.SG.M sleep-SC.3.SG.M
Intended: "Khalid was falling asleep."

(48) * χa:ləd ka:jɪn na:m
Khalid be.PSC.SG.M sleep-SC.3.SG.M
Intended: "Khalid was sleeping."

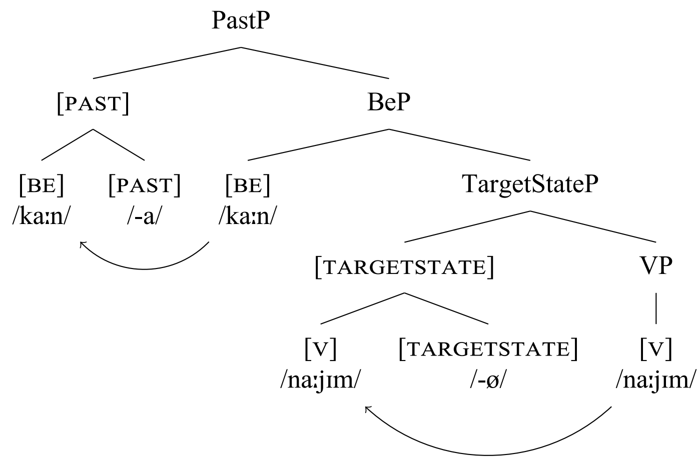
This supports the syntactic structure in (1), namely it reinforces the claim that the tense heads are higher than the [TARGETSTATE] and [PREPSTATE] heads. And just like the aspectual heads [EPISODIC] and [HABITUAL] which were realized on the stem of the higher auxiliary $\sqrt{\text{gʕd}}$ "sit", the tense conjugations only affix to the higher auxiliary root $\sqrt{kwn}$ in (45) and (46). I provide tree (49) as an illustration for a present tense target state sentence, and tree (50) for sentence (46)[28].

(49)



[28]Phonological note: As shown in Table 1, the PSC in the 3.SG.M seems to be null, but other phi features have clear exponents. The SC also seems to be null for 3.SG.M, but there is morphophonological evidence that there is an underlying /-a/ suffix, similar historically to Classical Arabic /-a/, however, it is obscured by some vowel deletion rules.

(50)



In (49), the verbal stem head-moves into the head [TARGETSTATE], and since the head [PRESENT] is phonologically null, there is no need for any ”Be-support”. On the other hand, in (50) we have the exact same head-movement of the verbal stem to [TARGETSTATE], but this time there is a non-phonologically null head [PAST] whose exponent is morphologically supported by the auxiliary stem /kaːn/. As far as I am aware, unlike English be which is related to the progressive, the auxiliary root $\sqrt{kwn}$ ”be” seems to be ”dummy” in the sense that it does not have any semantics of its own. It is just there to support the realization of a conjugation. This is unlike the auxiliary root $\sqrt{\text{gʕd}}$ ”sit” which does have a semantics of its own as we will see in section §4.2.

3.6 Concluding Simple Sentences

In this section I have shown that all the conjugations that realize the heads in (1) may affix to the lexical stem. This is also evidence for their independence, as there are no elements such as adverbs or auxiliaries that co-occur with them (although the future prefix /b-/ is dubious in this respect). Thus whenever we have a single conjugation and a lexical stem, we have simple sentences. However, if there are multiple conjugations and only one lexical stem, the auxiliary stem $\sqrt{kwn}$ ”be” appears in order to support whatever conjugation that needs to be realized. I have also argued that there is a phonologically null tense head [PRESENT] in HA. This goes well with ”be-support”, as there is no need to support a phonologically empty head in the first place, unlike the phonologically non-empty heads [PAST] and [FUTURE]. More justifications are warranted for the head [EPISODIC], which I present in the next section.

4 The Auxiliary Root $\sqrt{\text{gʕd}}$ ”Sit”

In this section, I will elaborate more on the auxiliary root $\sqrt{\text{gʕd}}$ “sit”. §4.1 gives evidence on why it is indeed an auxiliary and not some sort of lexical co-predicate.

§4.2 elaborates on the "durative" semantics it yields. §4.3 defends the existence of the head [EPISODIC] using this auxiliary root as well as other criteria. Finally, §4.4 concludes.

4.1 A Full-Fledged AUX

In modern Arabic dialects, the root $\sqrt{\text{gʕd}}$ "sit" as an auxiliary is used in "progressive" contexts (iterative events, preparatory states, "lately" habituals such as "He's not taking his medicine lately"[29]). See Camilleri and Sadler (2020) (C&S) for a plethora of examples from various Arabic dialects. Evidence for treating this root as an auxiliary comes from two sets of data: Co-predication and inanimate subjects.

For co-predication, (C&S) cite some dialects where using the root $\sqrt{\text{gʕd}}$ with its stem affixed with the PSC along with a lexical root whose stem is affixed with the PC yield two readings, a lexical and a grammatical one. One example is (51) from Tunisian Arabic.

(51) wəħid ga:ʕid jə-kol
one sit.PSC.Sgm 3-eat.PC.SG.M
Lexical Reading: "Someone is sitting while eating."
Grammatical Reading: "Someone is eating." (Tunisian Arabic, Saddour (2009, P273))

In (51), there are two roots, $\sqrt{\text{gʕd}}$ and $\sqrt{\text{ʔkl}}$ "eat". $\sqrt{\text{gʕd}}$'s verbal stem is affixed with the PSC, while $\sqrt{\text{ʔkl}}$'s verbal stem is affixed with the PC. Two readings are available: (i) a lexical reading where both verbs are interpreted lexically (there is a sitting and an eating event) and (ii) a grammatical reading (there is an on-going eating event). Thus $\sqrt{\text{gʕd}}$ is used lexically in the former but not the latter.

However, (my) HA differs from (51). I only obtain a grammatical reading, as in (52). In order for me to replicate the lexical reading of (51), I would produce (53):

(52) χa:ləd ga:ʕəd ja-akəl
Khalid sit.PSC.SG.M 3-eat.PC.SG.M
Obtained: "Khalid is eating."
Unobtained: "Khalid is eating while sitting."

(53) χa:ləd ga:ʕəd ja-akəl ga:ʕəd
Khalid sit.PSC.SG.M 3-eat.PC.SG.M sit.PSC.SG.M
Obtained: "Khalid is eating while sitting."
Unobtained: "Khalid is eating."

(52) only yields a grammatical reading. In order for me to obtain a lexical one, amusingly, I have to use the verb ga:ʕəd again, and thus I end up with two verbs whose

[29] I mention these points briefly in order to show how $\sqrt{\text{gʕd}}$ is similar crosslinguistically to other "progressive" morphemes. See Hughes and McClure (2017) and the resources there for Japanese's -teiru form that is similar to $\sqrt{\text{gʕd}}$ in being used for "progressive" as well as "lately/medal habitual tense". See also Starke (2022) and Dikmen and Demirok (2025) for crosslinguistic syncretism between this "medial habitual tense" and the "progressive".

stems are affixed with the PSC: The higher one which is an auxiliary, and a lower one which is lexical and modifies the manner of eating. Evidence that the lower one is a modifier can be shown by turning it into a manner interrogative word keːf ”how”. (54) is a question and (55) is its reply:

(54) keːf χaːləd gaːʕəd ja-akəl ?
how Khalid sit.PSC.SG.M 3-eat.PC.SG.M
”How/in what manner is Khalid eating?”

(55) χaːləd gaːʕəd ja-akəl waːgəf / gaːʕəd / mənbətˤəħ
Khalid sit.PSC.SG.M 3-eat.PC.SG.M stand.PSC.SG.M / sit.PSC.SG.M / lie.PSC.SG.M
”Khalid is eating while standing/sitting/laying (on his stomache/back)

There is further evidence that shows that the higher root $\sqrt{\text{gʕd}}$ is an auxiliary. C&S note an important distinction between lexical $\sqrt{\text{gʕd}}$ and auxiliary $\sqrt{\text{gʕd}}$: The former can not be used with inanimates, but the latter can. For example, in order for me to translate the English sentence ”The doll is sitting on the chair.”, I would need to use the root $\sqrt{\text{wd͡ʒd}}$ ”exists” instead of $\sqrt{\text{gʕd}}$:

(56) # ʔəd-dumj-ah gaːʕd-ah ʕal-al-kəɹsi
DEF-doll-SG.F sit.PSC-SG.F on-DEF-chair
Intended: ”The doll is sitting on the chair.”

(57) ʔəd-dumj-ah mawd͡ʒuud-ah ʕal-al-kəɹsi
DEF-doll-SG.F exist.PSC-SG.F on-DEF-chair
”The doll is sitting/exists on the chair.”

Using the root $\sqrt{\text{gʕd}}$ in (56) yields an infelicitious sentence. My fix is to use (57) with the root $\sqrt{\text{wd͡ʒd}}$. It appears that in many Arabic dialects, including mine, posture verbs are reserved for animate subjects. However, there are contexts where $\sqrt{\text{gʕd}}$ can be used with the inanimate subject ”doll”. Assuming someone set the doll on fire, I can produce (58):

(58) ʔəd-dumj-ah gaːʕd-ah tə-ħtəɹəg
DEF-doll-SG.F sit.PSC-SG.F 3.SG.F-burn.PC
”The doll is burning.”

In (58), the subject is still an inanimate ”the doll”, but $\sqrt{\text{gʕd}}$ can be used. The fact that the root $\sqrt{\text{gʕd}}$ can be used in (58) but not in (56) can be explained if we take the root to be an auxiliary in (58) but not (56).

In conclusion, one can know whether the root $\sqrt{\text{gʕd}}$ is an auxiliary in HA or not from its distribution and semantics. The auxiliary root is higher than the matrix lexical verb, no lexical reading is obtained as in (52), unless there is another lexical co-predicate, and it is compatible with inanimate subjects as a lexical root.

4.2 Durative Semantics

Alotaibi (2019) cites Al-Najjar (1984)'s use of the verb gaʕad "sit.SC" in Kuwaiti Arabic as a "durative aspectual verb". She continues: "Durativity includes progressive events and continuous states." Alotaibi (2019, P159). The usage of "durativity" here is too broad. For the sake of this paper, I will take the term "durative" to mean "a contextually lengthy period of the eventuality". This will explain nicely $\sqrt{\text{gʕd}}$'s incompatibility with the "on the verge of ending" reading with the simple [PREPSTATE] (see §3.1), and also its incompatibility with target state readings that have just begun, as in (59):

Context: Khalid has just fallen asleep. Abdullah points at him and says:

(59) # χaːləd gaːʕəd naːjɪm
Khalid sit.PSC.Sgm sleep.PSC.SG.M
Intended: "Khalid is sleeping."

(59) is infelicitious unless a contextual amount of time passes. Both the "on the verge of ending" contexts and (59), which is a state that has just been obtained, do not allow $\sqrt{\text{gʕd}}$. Based off this incompatibility, I take the auxiliary root $\sqrt{\text{gʕd}}$ to be the realization of the head [SIT] which denotes this durative semantics.

This should suffice for the current paper. However, I need to point out that this is a huge simplification of the role of $\sqrt{\text{gʕd}}$. Personally, it seems that I use $\sqrt{\text{gʕd}}$ in a lot of contexts which would be translated with the verbs keep, stay, and remain. To illustrate an instance of this point, consider the degree achievement verb "cool" with the root $\sqrt{\text{bɹd}}$ used in the past:

(60) ʔəʃ-ʃoːɹb-ah gʕad-at tə-bɹəd saːʕ-ah
DEF-soup-SG.F sit.SC-SG.F 3.SG.F-cool.PC hour-SG.F
"The soup kept cooling / cooled for an hour."

In (60), the verbal stem of $\sqrt{\text{gʕd}}$ is affixed with the SC expressing [PAST] while the lexical stem is affixed with the PC expressing [PREPSTATE]. The importance of (60) comes from comparing it with another Hasawi speaker, who produced it as (61):

(61) ʔəʃ-ʃoːɹb-ah ðˤalː-at tə-bɹəd saːʕ-ah
DEF-soup-SG.F remain.SC-SG.F 3.SG.F-cool.PC hour-SG.F
"The soup kept cooling / cooled for an hour."

(61) is very similar to (60) with one difference: The auxiliary root is different. It is $\sqrt{\text{gʕd}}$ in (60), but $\sqrt{\text{ðˤl}}$ "stay/remain" in (61). The same speaker who produced (61) uses the auxiliary root $\sqrt{\text{gʕd}}$ similar to me in contexts such as on-going preparatory states, but he feels that using $\sqrt{\text{gʕd}}$ with (61) is bad. An important take-away is that my auxiliary root $\sqrt{\text{gʕd}}$ is used in various on-going contexts (preparatory state events, iterative events, or degree achievements), whereas other dialects/idiolects might employ different auxiliary roots (Jordanian Arabic has $\sqrt{\text{gʕd}}$ "sit", $\sqrt{\text{ðˤl}}$ "remain", and $\sqrt{\text{nzl}}$ "descend" for different aspectual readings of on-going events, all of which I rendered with $\sqrt{\text{gʕd}}$).

In conclusion, I take the auxiliary root to realize the head [SIT] which encodes durativity, a contextually lengthy amount of time. However, cross-dialectal data points that this is a huge simplification and that at least for my idiolect this auxiliary root could be morphologically syncretic for various aspectually different readings.

4.3 Justifying Episodic

In this section, I will defend the existence of the head [EPISODIC] by highlighting the benefit of postulating it, which is mainly capturing the competition between the conjugations realizing [EPISODIC] and [HABITUAL] on being affixed to the auxiliary stem of auxiliary $\sqrt{\text{gʕd}}$ (the competition on being the terminal node of AktP as in (1)). After showing this competition, I will argue against reducing [EPISODIC] to [TARGETSTATE]. Lastly, I will present some data from Bedouin Palestinian Arabic (BPA) spoken in the Negev area, as reported by Henkin (1992). I do so as I believe BPA is similar to HA in using the PSC for realizing the head [EPISODIC].

In order to see the competition between the heads [EPISODIC] and [HABITUAL], consider sentences (62) and (63):

(62) χaːləd gaːʕəd ji-naam
Khalid sit.PSC.SG.M 3-sleep.PC.SG.M
"Khalid is falling asleep."

(63) χaːləd ja-gʕəd ji-naam
Khalid 3-sit.PC.SG.M 3-sleep.PC.SG.M
"Khalid keeps falling."

In both sentences, there is a preparatory state of falling asleep because of the the head [PREPSTATE] realized by the PC affixed on the lexical verbs. On the other hand, both auxiliary stems have different conjugations affixed to them. Semantically, (62) denotes a single occurance of falling asleep, unlike the habitual occurance of it in (63). If the auxiliary stem in (63) is affixed by the PC realizing the head [HABITUAL], then postulating a head [EPISODIC] that is realized by the PSC and is the opposite semantically of the head [HABITUAL] (single occurance versus multiple occurances) is a nice way to capture why we have a single event in (62) but a habitual one in (63). If this is true, then it amounts to saying that in (62) the auxiliary verb has two operators operating on the lexical verb: The auxiliary root $\sqrt{\text{gʔd}}$ which denotes that the eventuality is contextually lengthy, and the conjugation realizing the [EPISODIC] which quantifies over the eventuality to denote a single occurance of it.

While the competition above might point to warranting the postulation of a head [EPISODIC], there is another possibility of reducing the [EPISODIC] to [TARGETSTATE]. In this view, the auxiliary verb in (62) is a complex that has a conjugation realizing the [TARGETSTATE] which operates on the auxiliary root $\sqrt{\text{gʔd}}$. Then, the whole auxiliary verb acts as an operator on the lexical verb. The availability of this [TARGETSTATE] to the auxiliary root shouldn't be surprising, as the root $\sqrt{\text{gʔd}}$ can have a lexical stem affixed with the [TARGETSTATE] (it patterns like "sleep" type verbs, as was discussed in §2.3), and the auxiliary root might have "inherited" this combination of root-stem-conjugation. If this hypothesis is correct (call it ”inheritance” hypothesis), then we can reduce the head [EPISODIC] to [TARGETSTATE], which would simplify the structure in (1) and give an explanation of why the auxiliary stem is affixed with a PSC.

However, there is evidence against the inheritance hypothesis. Consider the modal auxiliary root $\sqrt{\text{gdɹ}}$ ”can”. Attempting to use $\sqrt{\text{gdɹ}}$ lexically is very bad to me:

(64) */?? χaːləd gədaɹ ʕal-al-ʕəlb-ah
Khalid can-SC.3.SG.M on-DEF-jar-SG.F
Intended: ”Khalid was able (to open) the jar.”

(64) is bad, and what comes to my mind is the attempt of inserting a lexical verb in order to fix it. The imporant bit is the fact that, this auxiliary can work very similarly to $\sqrt{\text{gʕd}}$ above in (62) and (63). To see this, consider sentence (65) where one wants to remark that Khalid is capable of opening a jar.

(65) χaːləd ji-gdaɹ ji-ftaħ l-ʕəlb-ah
Khalid 3-can.PC.SG.M 3-open.PC.SG.M DEF-jar-SG.F
”Khalid can open the jar.”

Sentence (65) is a habitual/dispositional/generic statement about Khalid's ability of opening a jar. If, say, Khalid kept opening jars until his hands were swollen, then it is appropriate to state (66):

(66) χaːləd mob gaːdəɹ ji-ftaħ l-ʕəlb-ah
Khalid NEG can.PSC.SG.M 3-open.PC.SG.M DEF-jar-SG.F
”Khalid can't open the jar.”

There is no contradiction between (65) and (66): Khalid can open the jar, it is just the case that he is temporally unable to due to his hands being swollen. Thus (66) is similar to (62) above: Both contain an auxiliary root whose stem is affixed with a PSC, and in both cases we have a temporary (not a habitual/dispositional/generic) context. More importantly, this is problematic for the inheritance hypothesis above: How can it be possible that the auxiliary stem in (66) is affixed with a PSC realizing the head [TARGETSTATE], despite the fact that the root $\sqrt{\text{gdɹ}}$ can't be used as a lexical root, and therefore there is no lexical root providing the ”inheritance” to the PSC realizing the head [TARGETSTATE]? If on the other hand we postulate a head [EPISODIC], then this problem does not arise. On the contrary, we capture the fact that the auxiliary roots are independent of the conjugations (the PSC realizing EPISODIC doesn't care if its stem belongs to the auxiliary root $\sqrt{\text{gʕd}}$ or $\sqrt{\text{gdɹ}}$), and that both the auxiliary root and the conjugation seem to be operators on the lexical verb.

Lastly, I briefly mention data from Henkin (1992) about BPA. Henkin investigates the semantic contributions of the PSC in BPA. Both lexical and auxiliary roots (co-)occur in the PSC. Of the auxiliary roots investigated were $\sqrt{kwn}$ ”be”, $\sqrt{\text{sˤjɹ}}$ ”become”, and $\sqrt{\text{ðˤl}}$ ”keep/stay” (this root is similar to the one in (61) above). What is important about the BPA data that Henkin talks about, is the fact that in every sentence she reported where there was a PSC affixed on the auxiliary stem, no habitual/generic reading was ever reported. They were all temporary/episodic. I provide illustrative examples for each auxiliary root:[30].

(67) kaːjne kaːtbe haːda Kaːmle
be.PSC.SG.F write.PSC.SG.F this Kaːmle
"Kaamle must have written this out (at home)." Henkin (1992, P441)

(68) sˤaːjra ħibːiː-ha ?
become.PSC.3.SG.F love.PC.3.SG.F-her ?
"Have you taken a sudden liking to her?" Henkin (1992, P442)

(69) dˤaːjl jəzaħlag
stay.PSC.SG.M slip.PC.3.SG.M
"(He) went skidding downhill (when he tripped...)." Henkin (1992, P442)

In (67), both the auxiliary and the lexical stems are affixed with a PSC. In (68) and (69), the auxiliary stem is affixed with a PSC, while the lexical stem is affixed with a PC. In all of these sentences, no habituality or genericity is reported. While Henkin (1992) did not talk about any episodic versus habitual/generic distinction as she was occupied with other categories (such as evidentiality), it seems HA is not unique in not obtaining a habitual/generic reading when the highest verbal stem is affixed with a PSC.

In conclusion, there seems to be a strong correlation between episodic semantics (taken here to mean a non habitual/generic/dispositional meaning) and the PSC affixed on auxiliary stems. As this conjugation can appear on the stem of any root ($\sqrt{\text{gʕd}}$ "sit" or $\sqrt{\text{gdɹ}}$ "can"), I take it to realize the head [EPISODIC] by its own which operates on the lexical verb instead of operating on the auxiliary root. This head can then be exploited for the stative roots that were introduced back in §3.2 (such as $\sqrt{\text{ħb}}$ "love" and $\sqrt{\text{kɹh}}$ "hate") as it can then account for why someone can love something, but hates it temporally, similar to (65) and (66) where someone could be capable of opening jars, but due to some circumstances such as swollen hands that person is temporally unable of doing so.

4.4 Wrapping up $\sqrt{\text{gʕd}}$

Concluding §4, I have shown why $\sqrt{\text{gʕd}}$ "sit" can be a lexical or an auxiliary (lexically bleached) root based off its distribution and semantic incompatibility with inanimate subjects. As an auxiliary, $\sqrt{\text{gʕd}}$ is unlike the auxiliary root $\sqrt{kwn}$ "be" as the latter is a "dummy" auxiliary introduced as a "be-support" operation in order to support the realization of a conjugation. $\sqrt{\text{gʕd}}$ on the other hand does indeed have semantics of its own. The semantics is related to prolonged eventualities and verbs such keep, stay, and remain. The auxiliary root $\sqrt{\text{gʕd}}$ was also used in order to justify the postulation of the head [EPISODIC], as the competition between the aspectual heads [HABITUAL] and [EPISODIC] can be shown when there is an auxiliary stem and

[30] Henkin (1992) does not gloss the examples, and she does not use the IPA. I have taken some liberties in glossing and converting her examples into IPA.

a lexical root. In the next section, I will elaborate more on the complex auxiliary sentences of HA, where both the auxiliary roots $\sqrt{\text{gʕd}}$ and $\sqrt{kwn}$ appear with a lexical root.

5 Double Auxiliary Structures

In this section, I will elaborate on complex AUX AUX V structures in HA. This requires elaborating on the syntactic restriction on lexical and auxiliary verbs' distributions, which forces the hierarchy of the verb BE c-commanding the verb SIT c-commanding the lexical verb V. I will also elaborate on the morphosyntactic restrictions on which verbal stem can conjugations affix to in these structures, similar to what I have done in §2.1 with SIT V structures and in §3.5 with BE V structures. The conclusion will be the justification of the syntactic hierarchy in (1).

The first point to elaborate on is the fact that HA is an AUX V[31] language. This is unlike its cousin Amharic which is a V AUX language Amberber (2010). Consider example (60) again, which has the structure SIT V. Switching the verbs around makes it ungrammatical, as in (70). This also happens with the "dummy" auxiliary $\sqrt{kwn}$ as in (71):

(70) * ʔəʃ-ʃoːɹb-ah tə-bɹəd gʕad-at saːʕ-ah
DEF-soup-SG.F 3.F.SG-cool.PC sit.SC-SG.F hour-SG.F
Intended: "The soup kept cooling / cooled for an hour."

(71) * χaːləd naːjɪm kaːn
Khalid sleep.PSC.SG.M be-SC.3.SG.M
Intended: "Khalid was sleeping."

Data such as (70) and (71) rule out any possible V AUX structures in HA. I have yet to come across any V AUX structures in my dialect.[32]

Turning to double auxiliary structures, both auxiliary roots $\sqrt{\text{gʕd}}$ and $\sqrt{kwn}$ can co-occur. When they do, we obtain a sentence such as (72):

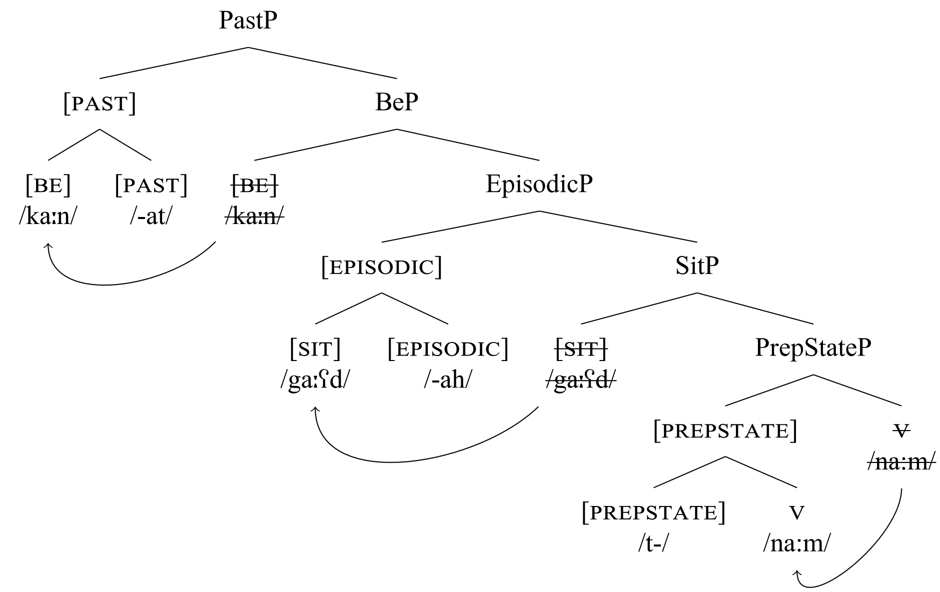
(72) maɹjam kaːn-at gaːʕd-ah t-naːm
Mary be-SC.3.SG.F sit-PSC.SG.F 3.SG.F-sleep.PC
"Mary was falling asleep."

In (72), ignoring the subject, there are three roots: The auxiliaries $\sqrt{\text{gʕd}}$ and $\sqrt{kwn}$, and the lexical $\sqrt{nwm}$. Interestingly, each verbal stem is affixed with a conjugation. The auxiliary stem of $\sqrt{kwn}$ is affixed with the SC realizing [PAST], $\sqrt{\text{gʕd}}$'s stem is affixed with the PSC realizing [EPISODIC], and $\sqrt{nwm}$'s stem is affixed with the PC realizing [PREPSTATE]. The whole sentence yields an episodic preparatory state event of falling asleep that is situated in the past. (73) illustrates the multiple stem-to-conjugation head-movements of (72):

[31] Again, I am ignoring here the problem of where the subject and object are generated. Argument structure is orthogonal as we obtain the same verbal and conjugational facts regardless of argument structure.

[32] There are cases where V AUX can surface, however it seems to be syntactically marked, as there needs to be a pause between V and AUX.

(73)



Syntactically, any deviation from this BE > SIT > V structure yields an ungrammatical sentence, regardless of the conjugations affixed to any verbal stem. This fact doesn't seem unique to HA. I have asked speakers of other dialects (Palestinian, Jordanian, Najdi, Lebanese, and Moroccan) whether they have the auxiliary root $\sqrt{kwn}$ below some other auxiliary root, and no one gave me an example of AUX BE V. With this, we can constrain the verbal root structure to the hierarchy as in (1).

$\sqrt{kwn}$'s c-commanding of other verbal stems can be explained by its "dummy" nature, as I have presented back in §3.5. (74) can illustrate this point further:

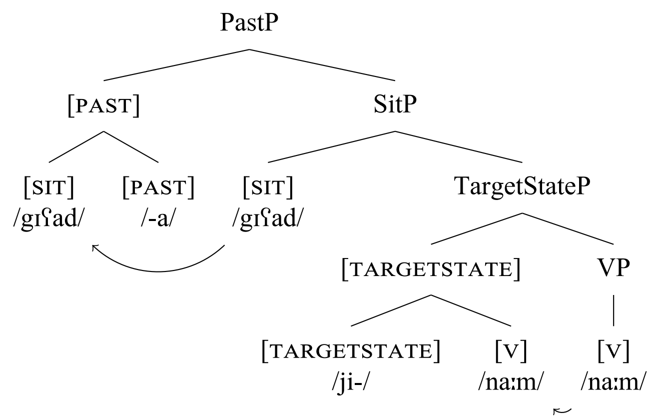
(74) χa:ləd gɪʕad ji-na:m
Khalid sit.SC.3.SG.M 3-sleep.PC.SG.M
"Khalid kept falling asleep / started falling asleep."[33]

In (74), there are two verbal roots, the auxiliary $\sqrt{\text{gʕd}}$ and the lexical $\sqrt{nwm}$. The former has its stem affixed with the SC realizing [PAST], while the latter has its stem affixed with the PC realizing [PREPSTATE]. I will take the absence of any PSC to be the absence of the heads [TARGETSTATE] or [EPISODIC]. If this is true, then we can account for the absence of the auxiliary root $\sqrt{kwn}$ from (74) easily: Because there is no [EPISODIC] head, and thus no PSC to realize said head, the auxiliary stem of $\sqrt{\text{gʕd}}$ is free to be affixed with any other conjugation. If [SIT] is merged after the lexical verb had been merged, and the next head to be merged is the head [PAST], then the SC realizing [PAST] will affix to the nearest free verbal stem, which happens to be

[33] I am unsure if this reading is ambiguous or underspecified between an iterative "He kept falling asleep" reading and an inceptive, episodic reading "He started falling asleep". A striking difference between (74) and (72) is that (72) can be used as an "overlap" reading (He was falling asleep when I entered on him), but not (74). As I have mentioned before, more semantic investigations are warranted for $\sqrt{\text{gʕd}}$.

$\sqrt{\text{gʕd}}$'s in (74). And so there is no need for "be-support" and the insertion of the root $\sqrt{kwn}$. In a way, one can say that the root $\sqrt{\text{gʕd}}$ "blocks" merging of the root $\sqrt{kwn}$.

(75)



Moving away from syntactic restrictions pertaining to the roots and to morphosyntactic restrictions pertaining to the conjugations, we will hear a similar story to the one we encountered back in §2.1 and §3.5. In multiple auxiliary sentences such as (72), only the PC realizing [PREPSTATE] and the PSC realizing [TARGETSTATE] can be affixed to the lexical stem. The auxiliary stem of $\sqrt{\text{gʕd}}$ can only be affixed with the PC realizing [HABITUAL] or the PSC realizing [EPISODIC]. Finally, as both the lexical stem of $\sqrt{nwm}$ and the auxiliary stem of $\sqrt{\text{gʕd}}$ already have conjugations affixed to them, the derivation is going to have a problem: The SC realizing [PAST] must affix to a stem, but both verbal stems are already affixed. This is where a last resort operation of "be-support" kicks in, and the root $\sqrt{kwn}$ is merged along with its verbal stem in order to support the SC. Similar thing will happen if the head [FUTURE] must be realized. However, this problem does not happen in present tense sentences, as the head [PRESENT] is phonologically null, and thus is why the present tense equivalent of (72) is exactly the same as (72) minus the auxiliary verb kaːn "was". In the appendix, I will provide a detailed list and examples of all the attested combinations in (1)

In conclusion, the verbal structure of HA is constrained in such a way that we obtain the structure in (1). The verbal roots are ordered in the sequence of BE > SIT > V, based off the fact that any deviation from this sequence yields an ungrammatical sentence, and off the fact that the auxiliary root $\sqrt{kwn}$ is a last resort operation that the grammar relies on in order to support the realization of the conjugations. The conjugations are also constrained as they can not affix to any verbal stem and they must affix locally to the nearest free verbal stem that they c-command.

6 Conclusion and Loose Ends

In this section, I will discuss short-comings and loose ends of the structure in (1) which I hope to return to in future work. §6.1 addresses the problem of conjugations having a one-to-many form to meaning relationship. §6.2 addresses a look-ahead problem with the "dummy" auxiliary root $\sqrt{kwn}$. §6.3 addresses two problematic cases of apparent double marking of tense morphology (co-occurring of past and future) and double marking of aspect (habitual and episodic co-occurance). Finally, and in anticipation to the data of the appendix, §6.4 discusses a gap of root and conjugation combination between the root $\sqrt{kwn}$ and the PSC.

6.1 Brute Forcing the Conjugations

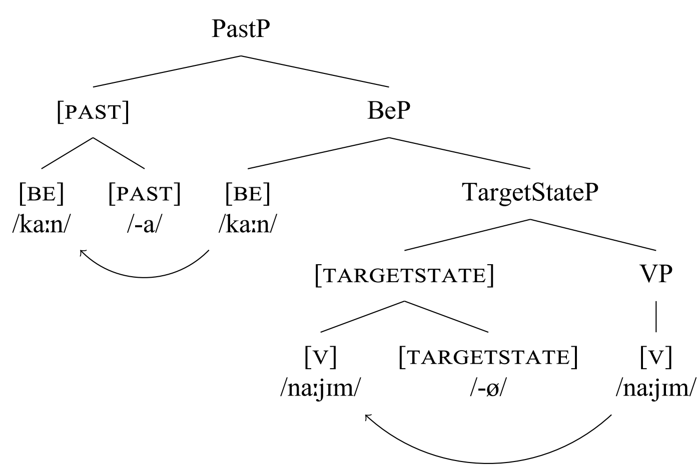
In this paper, I have not adopted any intricate morphological theory. A problem with this is the fact that the current proposal does not capture the fact that the three heads [PREPSTATE], [HABITUAL], and [FUTURE] form a natural class by being realized by the PC. Instead, the current proposal amounts to saying that there are three prefixal conjugations, and that they are all accidentally homophonous. In order to capture this complex relationshop between form and meaning, future work could consult morphosyntactic frameworks and their mechanisms. Contextual insertion of vocabulary items as in Distributed Morphology (Bobaljik (2017)), spanning (Svenonius (2012)), or phrasal lexicalization as in Nanosyntax (Baunaz et al. (2018)) are potential candidates for this task. Besides the heads that I have postulated here and their semantics, there remain many more that I have put aside due to space and time constraints (such as the perfect/anteriority relationship of event time before assertion time, which may be realized by the PSC or SC). I leave trying to figure out these semantics and how they fit with the structure in (1) to future work.

6.2 Last Resort and a Look Ahead Problem

Consider the past tense target state sentence (46) that we looked at back in §3.5. I replicate it and its syntactic representation below for convenience.

(76) χaːləd kaːn naːjɪm
Khalid be-SC.3.SG.M sleep.PSC.SG.M
"Khalid was sleeping."

(77)



The attentive reader might have noticed that there is a look-ahead problem with the "dummy" root $\sqrt{kwn}$. If the auxiliary root and stem combination of /kaːn/ (the combination of the root $\sqrt{kwn}$ and Form I's /CVCVC/ CV-skeleton) in (77) is licensed by the need to be affixed by the SC realizing the head [PAST], then how come do the auxiliary root and stem "know" that they will be needed before the head [PAST] is merged? I leave this issue while hoping of finding a solution in future work.

6.3 Double Tense Marking and Double Aspect Marking

There are two apparent problematic cases against the syntactic structure of (1). The first is an apparent co-occurance of two tense morphemes in the same sentence (co-occurance of a [SC] and the prefix /b-/). The second is an apparent co-occurance of both the aspectual heads [HABITUAL] and [EPISODIC]. I will elaborate more on the former set before the latter.

For tense morphemes, consider sentences (78) and (79) below:

(78) χaːləd kaːn b-j-aːkəl
Khalid be.SC.3.SG.M FUT-3-eat.PC.SG.M
"Khalid was about to eat."

(79) χaːləd b-i-kuːn ʔakal
Khalid FUT-3-be.PC.SG.M eat.SC.3.SG.M
"Khalid will have eaten."

In (78), the stem of the lexical root $\sqrt{\text{ʔkl}}$ "eat" is affixed with the PC which is prefixed with /b-/, while the stem of the auxiliary root $\sqrt{kwn}$ is affixed with the SC. In (79), on the other hand, we have the reverse situation with conjugations: The SC is affixed on the lexical stem, while the PC is prefixed on the auxiliary stem which is further prefixed by /b-/. This is problematic because we have both tense morphology appearing at the same time, and not only that, they appear to alternate between affixing on the lexical stem or the auxiliary stem.

However, upon closer scrutiny, these examples pose no problem for the syntactic structure in (1). They are another case of one-to-many relationship between form and meaning, namely, the SC and the prefix /b-/ are not just forms exclusive to past and future tense respectively. The following explanation is based off Fassi Fehri (1993, P147-148), who reports similar findings in Modern Standard Arabic (MSA) with sentences that have both past and future morphology co-occurring. The explanation goes like this: When two "tense" morphology co-occur, notice the fact that the tense interpretation of the sentence depends on what morphology the highest verb has: The auxiliary stem in (78) is affixed with the SC, and the reading that obtains is a past one. On the other hand, the auxiliary verb in (79) is prefixed with /b-/, and the reading that obtains is a future one. On the other hand, the morphology on the lower/lexical stem does not denote tense, but aspect. The "future" prefix /b-/ is on the lexical stem in (78), and the reading that obtains is a past prospective "was about to eat" reading (in Reichenbachian terms, assertion time is before event time, and both are before utterance time, hence AT < ET < UT). Sentence (79) on the other hand, has the lower/lexical stem affixed with the SC, and the obtained reading is one of the future perfect ”will have eaten” (UT < ET < AT). Thus HA is exactly as how Fassi Fehri (1993, P147-148) reported for MSA in these sentences where apparent tense morphology co-occur: Tense (the relationship between UT and AT) is denoted by the morphology of the auxiliary verb, while aspect (the relationship between AT and ET) is denoted by the morphology of the lower/lexical verb.

The conclusion is that (78) and (79) are not problematic for the structure in (1): The SC realizing [PAST] and the prefix /b-/ realizing [FUTURE] still compete on the terminal node of TP. However, future work should focus on pinpointing the structure of the lower aspectual heads that denote the relationship between AT and ET and apparently share the same morphology as the tense heads.

Next, I move to a more serious problem for the syntactic structure in (1), namely, the co-occurance of both the aspectual heads [HABITUAL] and [episodic]. Up to now, I have claimed that they are duals of each other: The head [HABITUAL] denotes habitual/dispositional/generic eventualities, while [EPISODIC] denotes a single occurance of these eventualities. I have tried to exploit this duality and claim that they compete for the same terminal node of AspP. However, consider the following problematic sentence from Maltese Arabic, as reported by Camilleri and Sadler (2020, P5):

(80) kon-t in-kun qiegħed ni-kteb u hu ji-ġi
be.SC-1SG 1-be.PC.SG sit.PSC.SG.M 1-write.PC.SG and he 3.M-come.PC.SG
j-waqqaf-ni
3.M-CAUS.stop.PC.SG-me
”I used to be writing, and he would come to stop me.”

(80) is, as (C&S) put it, a ”past habitual progressive”. Let us ignore the lower ”and” clause, as it is orthogonal and is only there to help facilitate the habitual reading. What is important is the main clause, where we have four verbs: Two ”be” verbs, one ”sit” verb, and a lexical ”write” verb. The crucial point lies with the fact that both the PC realizing [HABITUAL] and the PSC realizing [EPISODIC] co-occur. Since my paper is about my dialect, HA, this wouldn’t mean anything as I haven’t been discussing Maltese, and both dialects may vary greatly in their morphosyntax, thus my assessment in that these Maltese conjugations are similar to the conjugations of my dialect could be very misleading. However, the problem is the fact that I can replicate (80) very closely:

(81) kən-t ʔa-kuːn gaːʕəd ʔa-ktəb kəl maː ji-d͡ʒiː-ni
be.SC-1SG 1SG-be.PC sit.PSC.SG.M 1SG-write.PC each that 3-come.PC-me
”I used to be writing whenever he would come to me.”

Similar to Maltese, there are four verbs with the same roots, and all of them have the same conjugational patterns. The reading is also very similar: A past habitual progressive (on-going preparatory state). This sentence is very problematic for the view I have been pushing so far: How can we have both a head that denotes episodic eventualities and another which denotes habitual/dispositional/generic eventualities?

At this point, two options come to mind. Either that the PSC on the auxiliary stem ”sit” does not realize a head such as the [EPISODIC], or there might be a way to salvage the situation. I have currently no alternative proposal that replaces the head [EPISODIC]. However, there might be a very nice semantic explanation that salvages the situation. Instead of thinking of the co-occurance of both [EPISODIC] and [HABITUAL] as a problem, it could be seen in good light. Suppose that habitual eventualities are built out of episodic eventualities. If this is true, then HA and Maltese (assuming that the root and conjugation facts are similar) will be nice evidence for this hypothesis. Ferreira (2016) claims that the habitual and progressive share similar semantic concepts, but they differ in that the former is a plurality of eventualities while the latter is a single eventuality. If the semantics can be changed in such a way that one is derived from the other, then HA could be an instance of this derivation appearing on the surface (namely, the habitual is built on top of the episodic).[34]

Regardless of the status of the PSC and what it realizes in sentences such as (81), it seems that there is no competition between the head [HABITUAL] and whatever head ends up being realized by the PSC: Their co-occurance and the fact that the auxiliary verb denoting habitual is closer to the auxiliary verb denoting past tense suggests that they both are merged independently, with the head [HABITUAL] being higher than whatever head ends up being realized by the PSC.

On a side note, notice that in both Maltese and Hasawi we have also a co-occurance of two auxiliary roots $\sqrt{kwn}$. This can be taken as a reinforcement to the claim that this auxiliary root is "dummy" in that it does not have a semantics of its own. There are two non-dummy roots in (81) (lexical $\sqrt{\text{ʔkl}}$ and auxiliary $\sqrt{\text{gʕd}}$) and four conjugations. The lexical stem is affixed with the PC realizing [PREPSTATE], while the auxiliary stem of $\sqrt{\text{gʕð}}$ is affixed with the PSC (presumably) realizing [EPISODIC]. Since there are two more conjugations left to be supported (the PC realizing HABITUAL and the SC realizing [PAST]) "be-support" kicks in twice and in the order predicted by (1) as the head [HABITUAL] is below the head [PAST]. This means that we have two look-ahead problems (and also the problem of (1) not showing double $\sqrt{kwn}$).

6.4 A Gap in the Paradigm

In anticipation to the patterns of the appendix, in this section, I will highlight a problematic gap that the current theory predicts, and yet it does not surface. The current theory predicts that, whenever there are more conjugations than there are roots, as discussed in §3.5, the dummy auxiliary root $\sqrt{\text{kwn}}$ appears with a stem in order to support the realization of the conjugation. So far, we have seen this happen with the tense heads [PAST] and [FUTURE], as well as the aspectual head [HABITUAL] (double dummy support cases are attested as discussed with the Maltese inspired example (81) above). Thus we predict cases where we find a morphosyntactic combination of, say, a lexical stem, and two conjugations, the PC realizing [PREPSTATE] and the PSC realizing [EPISODIC]. In this case, the PC realizing [PREPSTATE] should affix to the lexical stem, while be-support appears in order to support the realization of the PSC realizing the [EPISODIC].

However, there is as of yet no case of the PSC realizing the head [EPISODIC] being affixed to the auxiliary stem of $\sqrt{kwn}$. In fact, at least for HA (and according to

[34] I am grateful to Hamida Demirdache for pointing me to Marcelo Ferreira's work and making me consider this option.

consultants of nearby dialects such as Dammami Arabic) there is no be.PSC verb at all. The putative verb *ka:jɪn is very bad to my ears and the ears of other Hasawi speakers. The closest that comes to rendering this root into be.PSC is using the Classical Arabic version, ka:ʔin, which Hasawi speakers would use for the noun "creature" [35]. Compare this to the Bedouin Palestinian Arabic data as presented in §4.3, where sentence (67) clearly has a PSC affixed on the auxiliary stem of the root $\sqrt{kwn}$. This never happens in HA, however. Despite my claims that conjugations do not care about the root-stem combination that they affix to (essentially, they should be blind to such a combination), I have no explanation for the absence of a putative verb *ka:jɪn "be.PSC" other than to claim that this is a lexical gap.

A Attested and Unattested Patterns

In this appendix, I present a collection of examples for all of the attested and predicted combinations of roots (auxiliary or lexical) and conjugations, as well as the unattested and ruled out examples. The examples are ordered from simple cases (only lexical verbs, similar to §3) to gradually complex ones (AUX-BE/SIT V and then AUX-BE AUX-SIT V).

As I have elaborated on in §5, the only attested root ordering is BE > SIT > V, the other possible combinations are ruled out by (1). For the ordering of affixation of conjugations to stems, Table 2 below lists all of the attested and predicted examples and the number-link to each example below. Table 3 provides all of the ruled out, possible yet unattested roots and conjugations patterns. Note that a pattern such as BE.PC V.SC is morphologically attested, but not for a past habitual context. A SC can affix to the lower, lexical stem instead of the auxiliary stem, but it does not yield past tense semantics (see the discussion of double "tense" morphology in §6.3). For the problematic cases see the conclusion, namely the attested but unpredicted examples in §6.3, and the predicted but unattested examples in §6.4. For the examples below, I am using the lexical root $\sqrt{\text{wgf}}$ "to stand (up)"[36], but for episodic readings I am using $\sqrt{\text{kɹh}}$ for the reasons stated in §2.2.

Syntactic Heads	Pattern	Example
Present TargetState V	V.PSC	(82)
Present Episodic V	V.PSC	(83)
Present PreparatoryState V	V.PC	(84)
Present Habitual V	V.PC	(85)

[35] Notice how Hasawi speakers actually use this word in its Classical Arabic phonology: Hollow roots (roots that have the second consonant as a glide, $\sqrt{CGC}$ as in $\sqrt{kwn}$) change the glide into from /w/ into /j/ in Form I of the PSC in HA and other dialects. However, for Classical Arabic, the glides turn into a glottal stop. The conclusion is that, $\sqrt{kwn}$ can appear in Form I of the [PSC], yet it is only used as a noun, and this form never surfaces as a verb

[36] Because this root begins with a glide /w/, the 3.SG.F prefix /tə-/, monophthongization might occur, in which case the underlying form /təwgaf/ "stand.PC" turns into surface form [to:gaf]. Note also that the aforementioned prefix might have its vowel delete in context where consonant clusters are allowed. Thus the changes occurring on this prefix are purely phonological.

Future V	FUT-V.PC	(86)
Past V	V.SC	(87)
Present Habitual TargetState V	BE.PC V.PSC	(88)
Present Habitual PreparatoryState V	BE.PC V.PC	(89)
Future TargetState V	FUT-BE.PC V.PSC	(90)
Future Episodic V	FUT-BE.PC V.PSC	(91)
Future PreparatoryState V	FUT-BE.PC V.PC	(92)
Future Habitual V	FUT-BE.PC V.PC	(93)
Past TargetState V	BE.SC V.PSC	(94)
Past Episodic V	BE.SC V.PSC	(95)
Past PreparatoryState V	BE.SC V.PC	(96)
Past Habitual V	BE.SC V.PC	(97)
Present Episodic SIT TargetState V	SIT.PSC V.PSC	(98)
Present Episodic SIT PreparatoryState V	SIT.PSC V.PC	(99)
Present Habitual SIT TargetState V	SIT.PC V.PSC	(100)
Present Habitual SIT PreparatoryState V	SIT.PC V.PC	(101)
Future SIT TargetState	FUT-SIT.PC V.PSC	(102)
Future SIT PreparatoryState	FUT-SIT.PC V.PC	(103)
Past SIT TargetState	SIT.SC V.PSC	(104)
Past SIT PreparatoryState	SIT.SC V.PC	(105)
Future Episodic SIT TargetState	FUT-BE.PC SIT.PSC V.PSC	(106)
Future Episodic SIT PreparatoryState	FUT-BE.PC SIT.PSC V.PC	(107)
Future Habitual SIT TargetState	FUT-BE.PC SIT.PC V.PSC	(108)
Future Habitual SIT PreparatoryState	FUT-BE.PC SIT.PC V.PC	(109)
Past Episodic SIT TargetState	BE.SC SIT.PSC V.PSC	(110)
Past Episodic SIT PreparatoryState	BE.SC SIT.PSC V.PC	(111)
Past Habitual SIT TargetState	BE.SC SIT.PC V.PSC	(112)
Past Habitual SIT PreparatoryState	BE.SC SIT.PC V.PC	(113)

Table 2: Attested and predicted examples.

Syntactic Heads	Pattern
Present Habitual TargetState V	BE.PSC V.PC
Present Habitual PreparatoryState V	BE.PSC V.PC
Future TargetState V	BE.PSC FUT-V.PC
Future Episodic V	BE.PSC FUT-V.PC
Future PreparatoryState V	BE.PC FUT-V.PC
Future Habitual V	BE.PC FUT-V.PC
Past TargetState V	BE.PSC V.SC
Past Episodic V	BE.PSC V.SC
Past PreparatoryState V	BE.PC V.SC
Past Habitual V	BE.PC V.SC

Present Episodic SIT PreparatoryState V	SIT.PC V.PSC
Present Habitual SIT TargetState V	SIT.PSC V.PC
Future SIT TargetState	SIT.PSC FUT-V.PC
Future SIT PreparatoryState	SIT.PC FUT-V.PC
Past SIT TargetState	SIT.PSC V.SC
Past SIT PreparatoryState	SIT.PC V.SC
Future Episodic SIT TargetState	BE.PSC FUT-SIT.PC V.PSC
Future Episodic SIT TargetState	BE.PSC SIT.PSC FUT-V.PC
Future Episodic SIT PreparatoryState	FUT-BE.PC SIT.PC V.PSC
Future Episodic SIT PreparatoryState	BE.PSC FUT-SIT.PC V.PC
Future Episodic SIT PreparatoryState	BE.PSC SIT.PC FUT-V.PC
Future Episodic SIT PreparatoryState	BE.PC FUT-SIT.PC V.PSC
Future Episodic SIT PreparatoryState	BE.PC SIT.PSC FUT-V.PC
Future Habitual SIT TargetState	FUT-BE.PC SIT.PSC V.PC
Future Habitual SIT TargetState	BE.PC FUT-SIT.PC V.PSC
Future Habitual SIT TargetState	BE.PC SIT.PSC FUT-V.PC
Future Habitual SIT TargetState	BE.PSC FUT-SIT.PC V.PSC
Future Habitual SIT TargetState	BE.PSC SIT.PSC FUT-V.PC
Future Habitual SIT PreparatoryState	BE.PC FUT-SIT.PC V.PC
Future Habitual SIT PreparatoryState	BE.PC SIT.PC FUT-V.PC
Past Episodic SIT TargetState	BE.PSC SIT.SC V.PSC
Past Episodic SIT TargetState	BE.PSC SIT.PSC V.SC
Past Episodic SIT PreparatoryState	BE.SC SIT.PC V.PSC
Past Episodic SIT PreparatoryState	BE.PSC SIT.SC V.PC
Past Episodic SIT PreparatoryState	BE.PSC SIT.PC V.SC
Past Episodic SIT PreparatoryState	BE.PC SIT.SC V.PSC
Past Episodic SIT PreparatoryState	BE.PC SIT.PSC V.SC
Past Habitual SIT TargetState	BE.SC SIT.PSC V.PC
Past Habitual SIT TargetState	BE.PSC SIT.SC V.PC
Past Habitual SIT TargetState	BE.PSC SIT.PC V.SC
Past Habitual SIT TargetState	BE.PC SIT.SC V.PSC
Past Habitual SIT TargetState	BE.PC SIT.PSC V.SC
Past Habitual SIT PreparatoryState	BE.PC SIT.SC V.PC
Past Habitual SIT PreparatoryState	BE.PC SIT.PC V.SC

Table 3: Ruled out, unattested root and conjugation patterns.

A.1 V Sentences

(82) maɹjam waːgf-ah
Mary stand.PSC-SG.F
"Mary is standing." (Present Target State)

(83) maɹjam kaːɹh-ah s-salamon
Mary hate.PSC-SG.F DEF-salmon

”Mary is hating the salmon.” (Present Episodic)

(84) maɹjam t-oːgaf
Mary 3.SG.F-stand.PC
”Mary is standing up.” (Present Preparatory State)

(85) maɹjam t-oːgaf
Mary 3.SG.F-stand.PC
”Mary stands up.” (Present Habitual)

(86) maɹjam b-t-oːgaf
Mary FUT-3.SG.F-stand.PC
”Mary will stand up.” (Future)

(87) maɹjam wəgəf-at
Mary stand.SC-3.SG.F
”Mary stood up.” (Past)

A.2 AUX V Sentences

A.2.1 BE V Sentences

(88) maɹjam t-kuːn waːgf-ah
Mary 3.SG.F-be.PC stand.PSC-SG.F
”Mary is standing.” (Present Habitual Target State[37])

(89) maɹjam t-kuːn t-oːgaf
Mary 3.SG.F-be.PC 3.SG.F-stand.PC
”Mary stands up.” (Present Habitual Preparatory State[38])

(90) maɹjam bə-t-kuːn waːgf-ah
Mary FUT-3.SG.F-be.PC stand.PSC-SG.F
”Mary will be standing.” (Future Target State)

(91) maɹjam bə-t-kuːn kaːɹh-ah l-salmon
Mary FUT-3.SG.F-be.PC hate.PSC-SG.F DEF-salmon
”Mary will be hating the salmon.” (Future Episodic)

(92) maɹjam bə-t-kuːn t-oːgaf
Mary FUT-3.SG.F-be.PC 3.SG.F-stand.PC
”Mary will be standing up.” (Future Preparatory State)

(93) maɹjam bə-t-kuːn t-oːgaf
Mary FUT-3.SG.F-be.PC 3.SG.F-stand.PC

[37] This sentence feels slightly different from (82) in that it is a present habitual target state reading, as in, ”Usually at this time, Mary is standing”.

[38] I am unsure how this sentence differs from (85). Both are habitual, but I can not discern any different semantic nuances between them so far.

"Mary will stand / be standing up." (Future Habitual[39])

(94) maɹjam kaːn-at wa:gf-ah
Mary be.SC-3.SG.F stand.PSC-SG.F
"Mary was standing." (Past Target State)

(95) maɹjam kaːn-at kaːɹh-ah s-salmon
Mary be.SC-3.SG.F hate.PSC-SG.F DEF-salmon
"Mary was hating the salmon." (Past Episodic)

(96) maɹjam kaːn-at t-oːgaf
Mary be.SC-3.SG.F 3.SG.F-stand.PC
"Mary was standing up." (Past Preparatory State)

(97) maɹjam kaːn-at t-oːgaf
Mary be.SC-3.SG.F 3.SG.F-stand.PC
"Mary used to stand up." (Past Habitual)

A.2.2 SIT V Sentences

(98) maɹjam gaːʕd-ah waːgf-ah
Mary sit.PSC.SG.F stand.PSC-SG.F
"Mary is standing." (Present Episodic Durative Target State)

(99) maɹjam gaːʕd-ah t-oːgaf
Mary sit.PSC.SG.F 3.SG.F-stand.PC
"Mary was standing up." (Present Episodic Durative Preparatory State)

(100) maɹjam ta-gʕəd waːgf-ah
Mary 3.SG.F-sit.PC stand.PSC-SG.F
"Mary keeps standing." (Present Habitual Durative Target State)

(101) maɹjam ta-gʕəd t-oːgaf
Mary 3.SG.F-sit.PC 3.SG.F-stand.PC
"Mary keeps standing up." (Present Habitual Durative Preparatory State)

(102) maɹjam b-ta-gʕəd waːgf-ah
Mary FUT-3.SG.F-sit.SC stand.PSC-SG.F
"Mary is standing." (Future Durative Target State)

(103) maɹjam b-ta-gʕəd t-oːgaf
Mary FUT-3.SG.F-sit.SC 3.SG.F-stand.PC
"Mary was standing up." (Future Durative Preparatory State)

[39] I am unsure of how to translate a future habitual in English. Native speakers told me that both "He will eat" and "He will be eating" could be used for a future habitual. Contrast this with the very clearly preferred past habitual "He used to eat". "He ate", on the other hand, doesn't give the implicature of past habituality.

(104) maɹjam gʕad-at wa:gf-ah
Mary sit.SC-3.SG.F stand.PSC-SG.F
"Mary kept standing." (Past Durative Target State)

(105) maɹjam gʕad-at t-o:gaf
Mary sit.SC-3.SG.F 3.SG.F-stand.PC
"Mary kept standing up." (Past Durative Preparatory State)

A.3 AUX AUX V Sentences

Note that these are the future/past counterparts to sentences (98)-(101).

(106) maɹjam bə-t-ku:n ga:ʕd-ah wa:gf-ah
Mary FUT-3.SG.F-be.PC sit.PSC-SG.F stand.PSC-SG.F
"Mary will remain standing." (Future Episodic Durative Target State)

(107) maɹjam bə-t-ku:n ga:ʕd-ah t-o:gaf
Mary FUT-3.SG.F-be.PC sit.PSC.SG-F 3.SG.F-stand.PC
"Mary will keep standing up." (Future Episodic Durative Preparatory State)

(108) maɹjam bə-t-ku:n ta-gʕəd wa:gf-ah
Mary FUT-3.SG.F-be.PC 3.SG.F-sit.PC stand.PSC-SG.F
"Mary will keep standing." (Future Habitual Durative Target State)

(109) maɹjam bə-t-ku:n ta-gʕəd t-o:gaf
Mary FUT-3.SG.F-be.PC 3.SG.F-sit.PC 3.SG.F-stand.PC
"Mary will keep standing up." (Future Habitual Durative Preparatory State)

(110) maɹjam ka:n-at ga:ʕd-ah wa:gf-ah
Mary be.SC-3.SG.F sit.PSC-SG.F stand.PSC-SG.F
"Mary remained standing." (Past Episodic Durative Target State)

(111) maɹjam ka:n-at ga:ʕd-ah t-o:gaf
Mary be.SC-3.SG.F sit.PSC-SG.F 3.SG.F-stand.PC
"Mary kept standing up." (Past Episodic Durative Preparatory State)

(112) maɹjam ka:n-at ta-gʕəd wa:gf-ah
Mary be.SC-3.SG.F 3.SG.F-sit.PC stand.PSC-SG.F
"Mary used to remain standing." (Past Habitual Durative Target State)

(113) maɹjam ka:n-at ta-gʕəd t-o:gaf
Mary be.SC-3.SG.F 3.SG.F-sit.PC 3.SG.F-stand.PC
"Mary used to keep standing up." (Past Habitual Durative Preparatory State)

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